

Global Supply Chain P&L

Power BI Web Application

Solution design for a secure, automated, standardised monthly SC P&L KPI application for OpCo / Country Supply Chain SPOCs and management - Actual vs Budget vs Last Year, in USD.

Item	Detail
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All figures shown in mock-ups, tables and the sample data set are SAMPLE / DUMMY data used only to illustrate structure.

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0. Source File Review - what the two reference files contain

Both reference files were found in the project folder and read in full (all sheets, cell formulas, data-validation lists, drawing shapes, embedded images and hyperlinks). No file was missing or unreadable, so the design proceeds. This section is the requested summary of findings, gaps and critical clarifying questions.

0.1 "Supply Chain P&L Dashboard - Template" (OUTPUT layout)

Tab	What it contains	Design implication
Dashboard Guide	Navigation tiles (Summary, Detailed, Trend Analysis). SMART account mapping list (Accounts Channel Accdetail), e.g. PURCH_PAIRS, PROD_PAIRS, PL_119001I ... PL_950000T, Channel = SC, Accdetail = F_CLO, plus 'Calculated' rows. General notes: amounts in USD; Unit / Thousand / Million scaling; blank = unmapped or n/a; (i) icon for KPI definitions; data sourced from SMART; refreshed Mon-Fri.	Account codes become Dim_Account. Channel and AccDetail are kept as fact attributes. A Home / Guide page reproduces the notes and the KPI glossary.
Summary	Title 'Supply Chain P&L Summary', subtitle 'Group - Total Business Unit - <Month Year>'. Left filter pane: Time Filter (Year 2025/2026, Month), Location Filter (Region: Africa, Asia-Pacific, Europe, India, Latam, Industrial; Country: 22 countries), Other Filter (BU: XYZ Ltd, AW Lab), Period (Monthly, Quarterly, YTD), Last Refresh Time. Unit / Thousand / Million toggle. Sections A-D, each with 3 KPI cards; card format = KPI name Monthly value, vs Bud, vs LY YTD value, vs Bud, vs LY.	Executive Summary page mirrors this layout exactly: 4 sections x 3 cards = 12 cards, same filters, same toggle. The template does not name the KPI per card (only 'KPI 1'), so the card allocation is an ASSUMPTION (A09).
Detailed - Supply Chain	P&L matrix with columns Actual Budget LY vs Bud % vs LY %. Rows: Purchase / Production Pairs, Turnover SC, Markup, Net Turnover, COGS, Inventory Valuation, Factory Personnel, Other Factory costs, Factory Costs, Recoveries, Factory (under)/over recovery, Freight, PPV, Cost of Sales, Total GP, GM %, Intercompany GP, Marketing, 4-Walls, Channel expenses (3 lines), Channel Expenses %, Channel Operating Profit / Margin, Enabling functions (3 lines), EBITDA, EBIT BNRI, Non Recurring Items, EBIT and margins. Sample formulas show how calculated rows are built. Function buttons link to 'Detailed - Imports / Local Sourcing / Manufacturing / Global Sourcing' (tabs not present in the file).	Detailed P&L page with a PL_Layout table and calculation groups for the 5 comparison columns; SC Function buttons replace the per-function tabs (same layout for every function).
Trend Analysis	Same filters (Month can be 'All'). Decomposition tree image (measure by Region -> OpCo -> Month), Actual vs Budget line trend by month, 'Cost Breakdown Tree Map' (Freight, Factory Personnel, Factory OH, Channel Expenses, PPV) and 'Cost Trend Analysis' (Freight, PPV, Factory Cost).	Trend Analysis page uses the same visuals, driven by a KPI selector. The 'ASP \$' in the sample images is not in the KPI file and is treated as illustrative only.

0.2 "Summary Page KPI's_Updated" (INPUT - KPI definitions)

Block	KPIs (as named in the file)	Formula given?
(A) Financial KPIs	Net Turnover Supply Chain; Turnover SC - Markup %; Gross Profit; Gross Margin %; EBITDA; EBITDA Margin %; EBIT; EBIT Margin %; Cost of Sales; Purchase Pairs; Production Pairs; Total Pairs	Descriptions only (e.g. 'SC Gross Earnings'); Markup % formula given. Other formulas are taken from the template's calculated rows.
Cost Analytics	Freight (Inbound) Cost per Pair; Factory Cost per Pair; Recovery per Pair; Cost of Sales (repeated); Channel expenses per pair; Channel Cost %; Cost of Sales %; Factory (under)/over recovery	Yes, except Channel Cost % (no formula; marked 'Existing').
B. Factory & Sourcing - Manufacturing	Factory Personnel Cost %; Other Factory Cost %	Yes. Remark: Turnover SC from the Manufacturing tab.
B. Factory & Sourcing - Procurement	PPV %; Import vs Local Mix; Average Cost per Pair (total, Import, LS, IHM)	Yes. Import vs Local Mix says 'Need formula', but the remark gives Import Mix % and Local Mix %.
C. Exception Management	Traffic lights: GP Margin decline >1% (Red); EBITDA negative (Red); Freight increase >10% (Amber); PPV adverse >5% (Amber); Recovery variance >10% (Red)	Thresholds given; the comparison base (vs Budget or vs LY) is not specified.

The 'Channel' column in the KPI file (All, Manufacturing, Import + LS, Import, LS, IHM) is the Supply Chain function scope of each KPI. It maps to the template's function tabs, and so to the SC Function dimension.

0.3 Gaps found

#	Gap / inconsistency	How it is handled
G1	Channel Cost % has no formula in the KPI file.	Template 'Channel Expenses %' = Channel Expenses / Net Turnover SC is used (A05).
G2	Import vs Local Mix: 'Need formula'.	Remark formula used: Import (LS) Purchase Pairs / Total Purchase Pairs.
G3	Cost of Sales % uses 'Turnover' with no further detail; Average Cost per Pair - LS uses 'LS COGS / Purchase'.	Net Turnover SC (A06) and LS Purchase Pairs (A10) are assumed.
G4	EBITDA / EBIT have descriptions only. The template sample sets EBITDA = GP + Channel Expenses and EBIT = EBIT BNRI - NRI, and ignores the Marketing, 4-Walls and Enabling Functions lines that sit in between.	EBITDA = Channel Operating Profit (SMART) + Enabling Functions (A07). EBIT = EBIT BNRI + NRI with SMART signs (A08). Both are open questions.
G5	Signs are inconsistent in the template sample: factory costs are positive, while freight and COGS are negative. Cost of Sales excludes Inventory Valuation.	Model convention: costs negative, as SMART delivers them (A04). Cost of Sales is taken as the SMART total PL_200000T.
G6	Several Guide account codes have no clear P&L line (PL_210000T, PL_220000T, OVERHEADS, PL_370000I, 970000, PL_940000T, PL_950000T).	Loaded and flagged 'TBC' in Dim_Account; they are not used in KPIs until confirmed (Q07).
G7	The per-function tabs (Imports, Local Sourcing, Manufacturing, Global Sourcing) are referenced but missing. The KPI file has no Global Sourcing KPIs.	Same P&L layout per function via the SC Function filter. Global Sourcing shows the P&L only (Q13).
G8	Traffic-light comparison bases are not defined.	Explicit rules assumed (A11) and configurable.
G9	Guide says 'refreshed daily (Mon-Fri)', but the brief asks for monthly publication.	Monthly published close; the refresh engine also supports daily refresh (Q08).
G10	No OpCo codes, SC Function column or currency column are visible in the template.	Extract column list assumed in 4.4 (A01/A02); to confirm with the SMART team.

0.4 Critical clarifying questions (design proceeds on stated assumptions)

The brief asks for questions and then for the design to proceed. The questions below are the ones that would change the design. Each has a working assumption so that work is not blocked. The full log is in Deliverable 14.

#	Question	Working assumption
Q01	Can SMART schedule a monthly extract (xlsx/csv) to a SharePoint / OneDrive folder, and with which columns?	Yes. Long-format file SCPL_YYYYMM_vN with the columns in 4.4.
Q02	Does the extract carry the SC Function (Imports / LS / Manufacturing / Global Sourcing) and BU on each row?	Yes (needed for Manufacturing / Procurement KPIs).
Q03	Is LY supplied by SMART (restated) or should it be derived from prior-year Actual?	Use SMART LY if present, else prior-year Actual.
Q04	Are amounts already in USD, and at which rate (actual, budget or constant)?	Delivered in USD by SMART; Currency must equal USD.
Q06	Confirm the EBITDA / EBIT build-up and signs (template shows EBIT = EBIT BNRI - NRI).	See G4 / A07 / A08.
Q09	Licensing: Power BI Pro for every viewer, or Premium / Fabric capacity?	Pro for the prototype; capacity recommended at scale.
Q10	Confirm the 12 Summary cards and the traffic-light comparison bases.	Allocation in 7.2 and rules in 5.4.

1. Business Problem and Business Impact

1.1 Problem statement

- **Limited visibility:** OpCo / Country Supply Chain SPOCs (Country SC Heads, IHM, Logistics, Quality, Sourcing and Finance Heads) cannot see their SC P&L KPIs when they need them.
- **No ERP access:** most SPOCs have no SMART report access and do not want to navigate the ERP for routine monitoring.
- **No standard:** there is no single, standardised global SC P&L dashboard. Each country receives different packs, definitions and layouts.

- **Manual Finance effort:** Global SC Finance spends significant time every month extracting, formatting, checking and emailing country information.
- **Management need:** leadership needs one consistent monthly view of Actual vs Budget vs Last Year across regions, countries, OpCos and SC functions, in USD.

1.2 Business impact of the current situation

Impact area	Effect today	Consequence
Decision speed	KPIs reach SPOCs late, after manual preparation (~WD8 or later, ASSUMPTION A15)	Corrective action on freight, PPV or factory absorption is delayed by a month
Consistency	Different definitions and layouts per country / pack	Comparisons across OpCos are unreliable; time is spent in reconciliation discussions
Finance capacity	Estimated ~53 hours per month of preparation and distribution (A15)	Controller time goes to reporting instead of analysis and business partnering
Control & security	Spreadsheets sent by email; no access control per country	Risk of sending the wrong data to the wrong person; no audit trail
Data quality	Checks are manual and undocumented	Errors can reach management; no visible data-quality status
Scalability	Each new OpCo, KPI or function adds manual work	Reporting cannot keep up with the business or with new analytics

2. Proposed Solution Overview

A single, secure, web-based **Power BI App - 'Global SC P&L'** publishes the standardised monthly Supply Chain P&L KPIs from SMART to authorised users, who need no ERP access. The data flow is automated from the SMART monthly extract to the user's email notification. Data is validated before it is published, and each user sees only their assigned Region / Country / OpCo / SC Function through dynamic Row-Level Security.

2.1 Key features

Feature	Description
Standard layout	Pages mirror the template: Home / Guide, Summary (Sections A-D, 12 KPI cards), Detailed P&L, Trend Analysis, plus OpCo / Country, IHM / Manufacturing and Sourcing pages.
KPI framework	28 KPIs, all taken from 'Summary Page KPI's_Updated', each with a documented definition, formula, DAX, source, frequency and unit.
Actual / Budget / LY	Every KPI can be shown as Actual, Budget, LY, Var vs Bud (abs / %) and Var vs LY (abs / %), for Monthly, Quarterly (QTD) and YTD. Two calculation groups provide this.
USD everywhere	All monetary values in USD, labelled on every card, chart and matrix. Unit / Thousand / Million toggle as in the template.
Automation first	SharePoint file drop, then Power Automate, Dataflow validation, Admin approval, refresh, and a personalised email. The Administrator does no manual dashboard preparation.
Data quality gate	15 validation rules (DQ01-DQ15). Critical errors stop publication; warnings need Administrator approval. A Data Quality & Refresh page is available to the Admin.
Security by design	Dynamic RLS driven by a User_Access SharePoint list, with no hard-coded users or countries. App audiences separate Viewers from the Administrator.
Self-service for viewers	Filter, multi-select, drill down / drill through, tooltips with KPI definitions, personal bookmarks, and export of permitted data.
History & scalability	Monthly history is retained (archive folder and model), so trends and YTD work from the first month. New OpCos, functions, KPIs or sources are added through master data, not through redesign.

2.2 Scope

In scope (prototype / MVP)	Out of scope (future roadmap)
SMART monthly extract (Actual, Budget, LY) for all SC functions Semantic model, 28 KPIs, 8 report pages, RLS	Direct SMART database / API connection Forecast scenario, predictive and AI features (Deliverable 13)

In scope (prototype / MVP)	Out of scope (future roadmap)
Validation, approval, refresh, email automation Admin user management and DQ monitoring	Local-currency / FX analysis Write-back of commentary

2.3 Design principles - how each is met

Principle	How the design meets it
Single source of truth (SMART)	Only SMART extracts feed the model. SMART totals (T accounts) are used as delivered and reconciled, not rebuilt.
Standardization	One KPI catalogue (Dim_KPI), one P&L layout (PL_Layout), one report used by every OpCo.
Automation first	Event-driven Power Automate flows and scheduled refresh. Manual steps are limited to the approval click and master-data maintenance.
User friendly	Business wording from the template, (i) KPI tooltips, navigation buttons, no ERP terms on pages.
Security by design	Entra ID sign-in, dynamic RLS, app audiences, no Build permission for viewers, sensitivity label 'Confidential'.
Scalable	Long (unpivoted) fact table, master-data-driven dimensions, measures generated by script, capacity-ready.
Management focused	Summary first, exceptions highlighted by traffic lights, drill to detail on demand.
Data quality before publication	Validation gate, publish flag per period (Publish_Control), DQ page and disclaimer in the email.
Auditability	Source file name and version on every fact row, file archive, exception log, refresh log, approval history.
USD reporting	Currency = USD enforced by rule DQ08. USD label and format string on every financial KPI.

3. Solution Architecture

Global SC P&L Power BI Application - End-to-end Solution Architecture

Automation-first monthly flow: SMART ERP to users, with a validation gate before publication and the Administrator in control

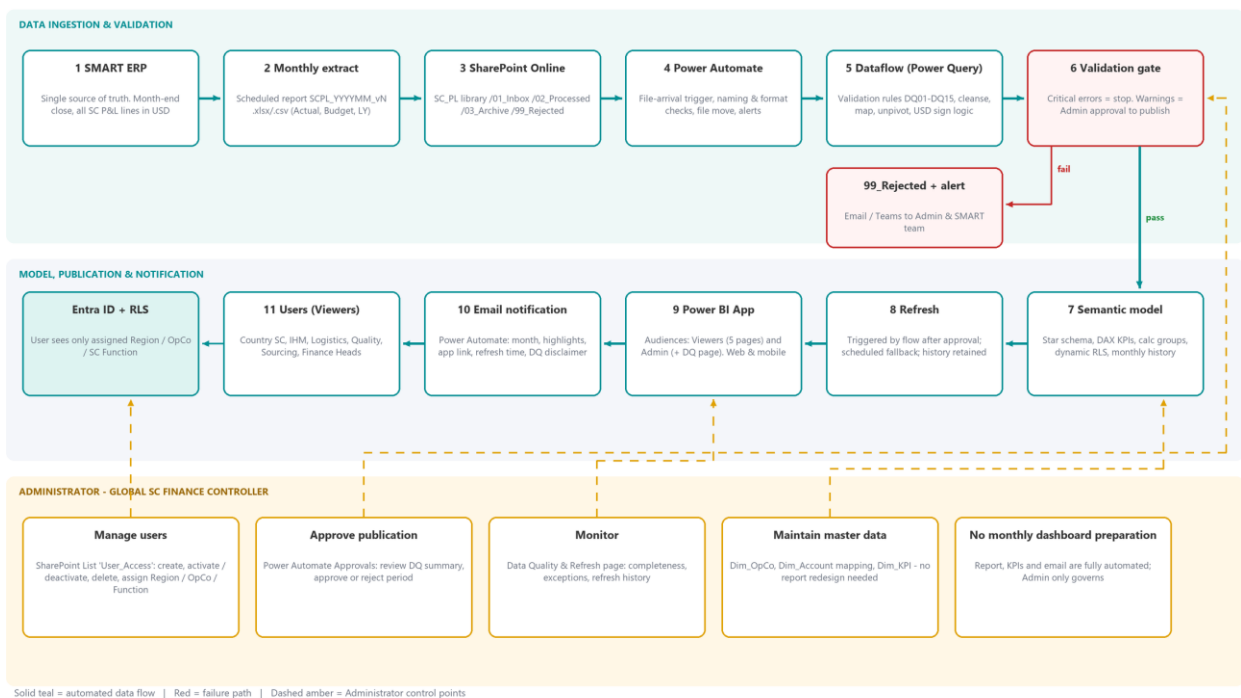


Figure 1 - End-to-end architecture and Administrator control points

3.1 End-to-end flow (Mermaid)

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flowchart LR
    A[SMART ERP] --> B[Scheduled monthly extract]
    B --> C[SCPL_YYYYMM_vN.xlsx / .csv]
  
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B --> C[SharePoint Online SC_PL/01_Inbox]
C -->|When a file is created| D[Power Automate SCPL-01 Intake]
D -->|Name / format check fails| R[99_Rejected + alert Admin and SMART team]
D --> E[Dataflow - Power Query validation DQ01-DQ15 and transformation]
E --> F{Critical errors?}
F -->|Yes| R
F -->|No| G[Approval request - Administrator]
G -->|Rejected| R
G -->|Approved| H[Publish_Control: period = Published file -> 03_Archive]
H --> I[Semantic model refresh star schema, DAX KPIs, RLS]
I --> J[Power BI App Viewer audience / Admin audience]
J --> K[Power Automate SCPL-02 monthly email]
K --> L[Authorised users - see only their RLS scope]
ADM((Administrator Global SC Finance Controller)) -->|reviews DQ page, approves| G
ADM -->|maintains User_Access list| L
ADM -->|monitors refresh and data quality| J
M[SCPL-03 Refresh monitor daily] -->|failure alert| ADM

```

3.2 Step-by-step flow and ownership

#	Step	Tool	Automated?	Owner
1	Month-end close; SC P&L lines posted in SMART, in USD	SMART ERP	n/a	OpCo Finance
2	Scheduled extract of the SC P&L report (Actual, Budget, LY) for the closed month	SMART scheduler / reporting	Yes	SMART / IT team
3	File lands in SharePoint SC_PL/01_Inbox with the naming convention	SharePoint Online (or OneDrive sync)	Yes	SMART / IT team
4	File-arrival trigger; name / format check; dataflow refresh	Power Automate (SCPL-01)	Yes	System
5	Validation rules DQ01-DQ15; cleansing, mapping, unpivot, sign and pairs split	Power BI Dataflow (Power Query)	Yes	System
6	Validation gate: critical errors reject the file; warnings go to approval	Power Automate + Approvals	Yes (decision by Admin)	Administrator
7	Period set to 'Published' in Publish_Control; file moved to 03_Archive	Power Automate + SharePoint List	Yes	System
8	Semantic model refresh (star schema, DAX, RLS); history retained	Power BI Service	Yes	System
9	App content available; RLS applied at query time	Power BI App	Yes	System
10	Monthly email: reporting month, highlights, link, refresh time, DQ disclaimer	Power Automate (SCPL-02) + Outlook	Yes	System
11	Users open the app, filter, drill, bookmark, export permitted data	Power BI web / mobile	n/a	Viewers
-	Daily refresh monitoring and missing-file check	Power Automate (SCPL-03) + Power BI REST API	Yes	System -> Admin

Administrator's role (Global SC Finance Controller). The Administrator governs the process but does not prepare dashboards:

- manages users and access in the User_Access list (create, activate / deactivate, delete, assign role, Region / OpCo / Function);
- reviews the Data Quality page and approves or rejects each month's publication;
- monitors refresh status and exceptions, and maintains master data (new OpCo, account mapping, KPI catalogue). Report design, KPI logic and the email are fully automated and change only through controlled releases.

4. Data Model

Semantic Model - Star Schema (Fact_SC_PL at month grain)

One-to-many (1 : *), single-direction filters from dimension to fact. Grey tables are disconnected (no relationship).

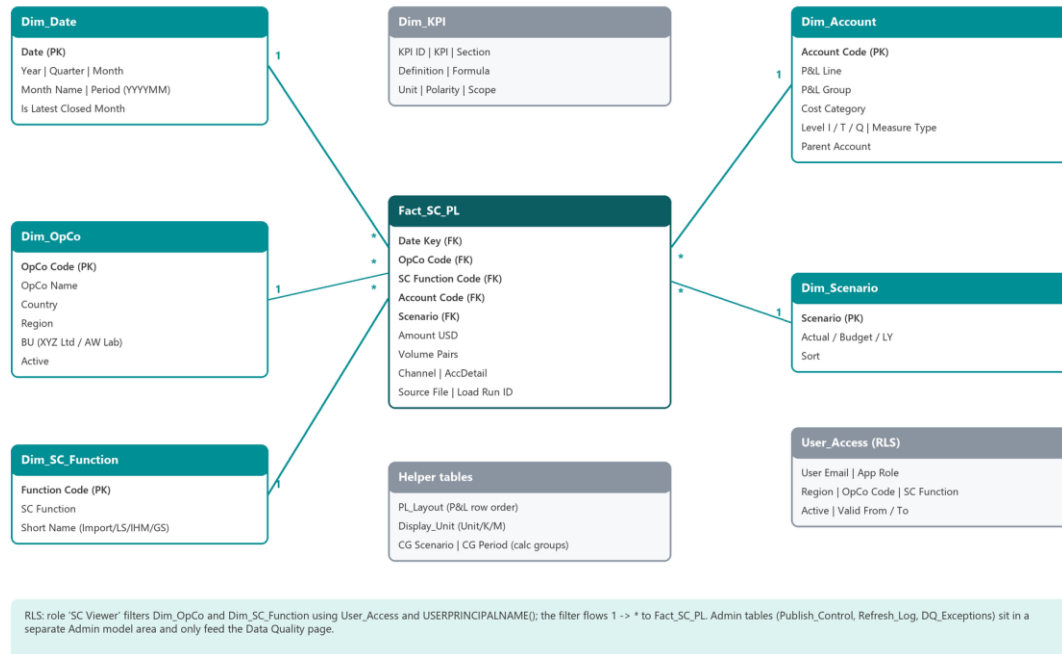


Figure 2 - Star schema of the semantic model

4.1 Tables

Table	Type	Grain / content	Key columns	Source
Fact_SC_PL	Fact	Month x OpCo x SC Function x Account x Scenario (x Channel, AccDetail)	Date Key, OpCo Code, SC Function Code, Account Code, Scenario, Amount USD, Volume Pairs, Source File, File Version	SMART monthly extract
Dim_Date	Dimension	Day (fact joins on 1st of month); Year > Quarter > Month hierarchy	Date, Year, Quarter, Month, Month Name, Period (YYYYMM), Period Selector	Power Query generated
Dim_OpCo	Dimension	OpCo -> Country -> Region, BU	OpCo Code, OpCo Name, Country, Region, BU, Active	Master data list (Admin)
Dim_SC_Function	Dimension	Imports, Local Sourcing, Manufacturing (IHM), Global Sourcing	Function Code, SC Function, Short Name, Sort	Master data
Dim_Account	Dimension (Cost Category)	SMART account (template Dashboard Guide codes)	Account Code, P&L Line, P&L Group, Cost Category, Level (I / T / Q), Measure Type (USD / Pairs), Parent Account, Mapping Status	Master data (Admin)
Dim_Scenario	Dimension	Actual, Budget, LY	Scenario, Sort	Static
Dim_KPI	Disconnected	One row per KPI (28)	KPI ID, KPI, Section, Description, Formula, Unit, Polarity, Channel Scope	KPI file
User_Access	Security (disconnected)	User x scope row	User Email, App Role, Region, OpCo Code, SC Function, Active, Valid From / To	SharePoint List (Admin)
PL_Layout / Display_Unit	Helper	P&L row order; Unit / K / M	Sort, Row Label, Row Type, Account / Measure, Format	Static
CG_Scenario / CG Period	Calculation groups	Actual, Budget, LY, Var vs Bud (%), Var vs LY (%) / Monthly, Quarterly, YTD	Comparison, Period	Model
Publish_Control, DQ_Exceptions, Refresh_Stamp	Admin / audit	Period status; validation issues; refresh time	Period, Status, Rule ID, Severity, Row No	SharePoint List / Power Query

The brief's KPI dimension is implemented as Dim_KPI, the KPI catalogue. It drives the KPI slicer, tooltips and the glossary. KPI values themselves are DAX measures, because ratios cannot be stored as additive facts. Cost Category is an attribute of Dim_Account.

4.2 Relationships and cardinality

From (one side)	To (many side)	Cardinality	Cross-filter	Note
Dim_Date[Date]	Fact_SC_PL[Date Key]	1 : *	Single	Dim_Date marked as date table
Dim_OpCo[OpCo Code]	Fact_SC_PL[OpCo Code]	1 : *	Single	RLS filter flows to the fact
Dim_SC_Function[Function Code]	Fact_SC_PL[SC Function Code]	1 : *	Single	RLS filter flows to the fact
Dim_Account[Account Code]	Fact_SC_PL[Account Code]	1 : *	Single	Cost Category / P&L group
Dim_Scenario[Scenario]	Fact_SC_PL[Scenario]	1 : *	Single	Used by the calculation groups
User_Access, Dim_KPI, PL_Layout, Display_Unit, Publish_Control	-	none	-	Disconnected on purpose (no ambiguity, no bi-directional filters)

4.3 Hierarchies and modelling rules

- Hierarchies: **Region > Country > OpCo; Year > Quarter > Month; P&L Group > P&L Line; SC Function.**
- Fact keys are hidden; users work only with dimension attributes and measures. Measures are grouped in display folders by KPI section.
- Signed amounts, so sums work across P&L lines (revenue +, costs -). Ratio and per-pair KPIs are calculated in DAX and never stored.
- History: each monthly file is archived and re-read, and only the highest version per period is used. Incremental refresh (by Date Key) is enabled when volumes grow.
- Only periods marked 'Published' by the Administrator enter Fact_SC_PL. Rows of unpublished periods remain visible only on the Admin DQ page.

4.4 Fact fields aligned to the input template

Extract column (assumed, A01)	Template reference	Model field	Rule
Year, Month	Summary 'Time Filter' (Year, Month)	Dim_Date via Date Key = 1st of month	Mandatory; must equal file period
Region, Country	Summary 'Location Filter'	Dim_OpCo[Region], [Country]	Mandatory; validated against Dim_OpCo
OpCo Code (+ OpCo Name)	Trend image 'Opco' level	Dim_OpCo[OpCo Code]	Mandatory
BU	Summary 'Other Filter' BU (XYZ Ltd, AW Lab)	Dim_OpCo[BU]	Attribute of OpCo (A13)
SC Function	Detailed buttons: Supply Chain, Imports, Local Sourcing, Manufacturing, Global Sourcing; KPI file 'Channel' column	Dim_SC_Function	Mandatory (A02)
Account (+ description)	Dashboard Guide 'Accounts' column	Dim_Account[Account Code]	Mapped; unmapped = DQ06
Channel	Dashboard Guide 'Channel' = SC	Fact attribute	Kept for audit
AccDetail	Dashboard Guide 'Accdetail' = F_CLO	Fact attribute	Kept for audit
Scenario	Detailed columns Actual, Budget, LY	Dim_Scenario	Actual / Budget / LY only
Currency	Guide note 'amounts in USD'	Validation only	Must be USD (DQ08)
Amount	Detailed values; PURCH_PAIRS / PROD_PAIRS rows hold pairs	Amount USD or Volume Pairs (split by Dim_Account[Measure Type])	Numeric; pairs >= 0 and whole numbers

5. KPI Framework

All KPI definitions below come from 'Summary Page KPI's_Updated' (column 'Description (source)'). Where the KPI file gives a formula it is used as written. Where it gives only a description, the formula is taken from the template's calculated rows and flagged as an ASSUMPTION. No KPI has been invented. Frequency: every KPI is published monthly and available as Monthly, Quarterly (QTD) and YTD. Currency: USD for amounts; per-pair KPIs are USD per pair.

ID	KPI / KPI-file section	Definition (from KPI file)	Formula	DAX measure (core)	Source	Frequency	Currency / polarity
K01	Net Turnover Supply Chain A. Financial KPIs	Total SC Revenue	Turnover Supply Chain + Turnover Supply Chain - Markup (per template)	[Net Turnover SC] = SMART PL_119000T	SMART PL_119000T (= PL_119001I + PL_119002I) Scope: All	Monthly; QTD; YTD	USD Higher is better
K02	Turnover Supply Chain - Markup % A. Financial KPIs	Turnover Supply Chain - Markup/Turnover Supply Chain	Turnover SC - Markup / Turnover SC	DIVIDE([Turnover SC Markup],[Turnover SC])	SMART PL_119002I, PL_119001I Scope: All	Monthly; QTD; YTD	% (of USD) Higher is better
K03	Gross Profit A. Financial KPIs	SC Gross Earnings	Net Turnover SC + Cost of Sales + Intercompany GP (per template)	[Gross Profit] = SMART PL_920000T	SMART PL_920000T (= Net Turnover + PL_200000T + PL_IC0000T) Scope: All	Monthly; QTD; YTD	USD Higher is better
K04	Gross Margin % A. Financial KPIs	GP % on Turnover	Gross Profit / Net Turnover SC	DIVIDE([Gross Profit],[Net Turnover SC])	SMART PL_920000T, PL_119000T Scope: All	Monthly; QTD; YTD	% (of USD) Higher is better
K05	EBITDA A. Financial KPIs	Operating Profitability	Channel Operating Profit + Enabling Functions Expenses (ASSUMPTION A07)	[Channel Operating Profit]+[Enabling Functions Expenses]	SMART PL_CP0000T, PL_OH0000T Scope: All	Monthly; QTD; YTD	USD Higher is better
K06	EBITDA Margin % A. Financial KPIs	EBITDA %	EBITDA / Net Turnover SC	DIVIDE([EBITDA],[Net Turnover SC])	SMART EBITDA, PL_119000T Scope: All	Monthly; QTD; YTD	% (of USD) Higher is better
K07	EBIT A. Financial KPIs	Final Operating Result	EBIT BNRI + Non Recurring Items (ASSUMPTION A08)	[EBIT BNRI]+[Non Recurring Items]	SMART EBITDA, PL_380000I (TBC) Scope: All	Monthly; QTD; YTD	USD Higher is better
K08	EBIT Margin % A. Financial KPIs	EBIT %	EBIT / Net Turnover SC	DIVIDE([EBIT],[Net Turnover SC])	SMART EBIT, PL_119000T Scope: All	Monthly; QTD; YTD	% (of USD) Higher is better
K09	Cost of Sales A. Financial KPIs	Total SC Cost excluding channel expenses	SMART account PL_200000T (COGS + under/over recovery + Freight + PPV)	CALCULATE([Amount USD], Account="PL_200000T")	SMART PL_200000T Scope: All	Monthly; QTD; YTD	USD Lower is better
K10	Purchase Pairs A. Financial KPIs	Procurement Volume	Sum of PURCH_PAIRS	CALCULATE([Pairs], Account="PURCH_PAIRS")	SMART PURCH_PAIRS Scope: Import + LS	Monthly; QTD; YTD	Pairs (volume) Higher is better
K11	Production Pairs A. Financial KPIs	Factory Output	Sum of PROD_PAIRS	CALCULATE([Pairs], Account="PROD_PAIRS")	SMART PROD_PAIRS Scope: Manufacturing	Monthly; QTD; YTD	Pairs (volume) Higher is better
K12	Total Pairs A. Financial KPIs	Total Volume	Purchase Pairs + Production Pairs	[Purchase Pairs]+[Production Pairs]	SMART PURCH_PAIRS, PROD_PAIRS Scope: All	Monthly; QTD; YTD	Pairs (volume) Higher is better
K13	Freight (Inbound) Cost per Pair Cost Analytics New KPI	Logistics productivity	Freight Costs / Purchase+Production Pairs	DIVIDE(-[Freight Costs],[Total Pairs])	SMART PL_225000I; pairs Scope: All	Monthly; QTD; YTD	USD / pair Lower is better
K14	Factory Cost per Pair Cost Analytics New KPI	Manufacturing efficiency	Factory Costs / Production Pairs	DIVIDE(-[Factory Costs],[Production Pairs])	SMART PL_227100T; PROD_PAIRS Scope: Manufacturing	Monthly; QTD; YTD	USD / pair Lower is better
K15	Recovery per Pair Cost Analytics New KPI	Manufacturing efficiency	Recoveries of Factory / Production Pairs	DIVIDE([Recoveries of Factory],[Production Pairs])	SMART PL_227600I; PROD_PAIRS Scope: Manufacturing	Monthly; QTD; YTD	USD / pair Higher is better
K16	Channel Expenses per Pair Cost Analytics New KPI	Selling efficiency	Channel Expenses / Purchase+Production Pairs	DIVIDE(-[Channel Expenses],[Total Pairs])	SMART PL_CH0000T; pairs Scope: All	Monthly; QTD; YTD	USD / pair Lower is better
K17	Channel Cost %	Selling efficiency	Channel Expenses / Net Turnover SC (per	DIVIDE(-[Channel Expenses],[Net Turnover SC])	SMART PL_CH0000T, PL_119000T	Monthly; QTD;	% (of USD)

ID	KPI / KPI-file section	Definition (from KPI file)	Formula	DAX measure (core)	Source	Frequency	Currency / polarity
	Cost Analytics Existing		template 'Channel Expenses %'; ASSUMPTION A05)		Scope: All	YTD	Lower is better
K18	Cost of Sales % Cost Analytics New KPI	Cost optimization	Cost of Sales / Turnover (Net Turnover SC; ASSUMPTION A06)	DIVIDE(-[Cost of Sales],[Net Turnover SC])	SMART PL_200000T, PL_119000T Scope: All	Monthly; QTD; YTD	% (of USD) Lower is better
K19	Factory (under)/over recovery Cost Analytics Existing	Factory absorption control	Recoveries of Factory - Factory Costs (SMART PL_227000T)	CALCULATE([Amount USD], Account="PL_227000T")	SMART PL_227000T (= PL_227600I + PL_227100T) Scope: Manufacturing	Monthly; QTD; YTD	USD Higher is better
K20	Factory Personnel Cost % B. Factory & Sourcing - Manufacturing New KPI	Labor efficiency	Factory Personnel / Turnover SC (Manufacturing)	CALCULATE(DIVIDE(-[Factory Personnel],[Turnover SC]), Function="Manufacturing")	SMART PL_227100I, PL_119001I (Mfg) Scope: Manufacturing	Monthly; QTD; YTD	% (of USD) Lower is better
K21	Other Factory Cost % B. Factory & Sourcing - Manufacturing New KPI	Overhead control	Factory OH / Turnover SC (Manufacturing)	CALCULATE(DIVIDE(-[Other Factory Costs],[Turnover SC]), Function="Manufacturing")	SMART PL_227200I, PL_119001I (Mfg) Scope: Manufacturing	Monthly; QTD; YTD	% (of USD) Lower is better
K22	PPV % B. Factory & Sourcing - Procurement New KPI	Purchase efficiency	PPV / Turnover SC (Imports + Local Sourcing)	CALCULATE(DIVIDE([PPV],[Turnover SC]), Function IN {Imports, LS})	SMART PL_224000I, PL_119001I (Imp + LS) Scope: Import + LS	Monthly; QTD; YTD	% (of USD) Higher is better
K23	Import Mix % B. Factory & Sourcing - Procurement New KPI	Sourcing strategy	Import Purchase Pairs / Total Purchase Pairs	DIVIDE(Import Purchase Pairs, Import+LS Purchase Pairs)	SMART PURCH_PAIRS by SC Function Scope: Import + LS	Monthly; QTD; YTD	% (of USD) n/a is better
K24	Local Sourcing Mix % B. Factory & Sourcing - Procurement New KPI	Sourcing strategy	Local sourcing Purchase Pairs / Total Purchase Pairs	DIVIDE(LS Purchase Pairs, Import+LS Purchase Pairs)	SMART PURCH_PAIRS by SC Function Scope: Import + LS	Monthly; QTD; YTD	% (of USD) n/a is better
K25	Average Cost per Pair B. Factory & Sourcing - Procurement New KPI	Procurement economics	COGS Supply chain / Purchase+Prod Pairs	DIVIDE(-[COGS SC],[Total Pairs])	SMART PL_219000T; pairs Scope: All	Monthly; QTD; YTD	USD / pair Lower is better
K26	Average Cost per Pair - Import B. Factory & Sourcing - Procurement	Procurement economics	Import COGS Supply chain / Purchase Pairs	CALCULATE(DIVIDE(-[COGS SC],[Purchase Pairs]), Imports)	SMART PL_219000T, PURCH_PAIRS (Imports) Scope: Import	Monthly; QTD; YTD	USD / pair Lower is better
K27	Average Cost per Pair - LS B. Factory & Sourcing - Procurement	Procurement economics	LS COGS Supply chain / Purchase (Pairs; ASSUMPTION A10)	CALCULATE(DIVIDE(-[COGS SC],[Purchase Pairs]), Local Sourcing)	SMART PL_219000T, PURCH_PAIRS (LS) Scope: LS	Monthly; QTD; YTD	USD / pair Lower is better
K28	Average Cost per Pair - IHM B. Factory & Sourcing - Procurement	Procurement economics	IHM COGS Supply chain / Prod pairs	CALCULATE(DIVIDE(-[COGS SC],[Production Pairs]), Manufacturing)	SMART PL_219000T, PROD_PAIRS (Mfg) Scope: IHM	Monthly; QTD; YTD	USD / pair Lower is better

5.1 Comparison and time-intelligence measures (apply to every KPI)

Measure	Definition	Formula	DAX (calculation-group item)	Frequency	Currency
Actual	Closed actuals from SMART	KPI evaluated for Scenario = Actual	CALCULATE(SELECTEDMEASURE(), Dim_Scenario[Scenario]="Actual")	Monthly	As KPI
Budget	Approved budget	KPI for Scenario = Budget	CALCULATE(SELECTEDMEASURE(), Dim_Scenario[Scenario]="Budget")	Monthly	As KPI
LY	Same period last year	SMART LY, else prior-year Actual	IF(NOT ISBLANK(LY), LY, CALCULATE(SELECTEDMEASURE(), Actual, DATEADD(Dim_Date[Date],-1,YEAR)))	Monthly	As KPI

Measure	Definition	Formula	DAX (calculation-group item)	Frequency	Currency
Variance vs Budget	Actual - Budget	A - B (ratio KPIs: percentage points)	_a - _b	Monthly	USD / pp
Variance vs Budget %	Relative variance	(A - B) / B ; ratio KPIs: (A - B) x 100 pp	IF(_isRatio, (_a-_b)*100, DIVIDE(_a-_b, ABS(_b)))	Monthly	%
Variance vs LY / vs LY %	Actual - LY	(A - LY) / LY	same pattern with LY logic	Monthly	USD / % / pp
QTD	Quarter to date	Sum of the months in the quarter up to the selected month	CALCULATE(SELECTEDMEASURE(), DATESQTD(Dim_Date[Date]))	Monthly	As KPI
YTD	Year to date (calendar year, A12)	Sum of Jan to the selected month; ratios recalculated on YTD components	CALCULATE(SELECTEDMEASURE(), DATESYTD(Dim_Date[Date]))	Monthly	As KPI

Variance display follows the template: 'n.a' when the base is blank (e.g. no LY). Colour follows polarity: for P&L amounts higher is better (because costs are negative); for per-pair and cost-% KPIs lower is better (Dim_KPI[Polarity]).

5.2 DAX measures - base and P&L lines

```
// Scenario-safe base: if no single scenario is in context, default to Actual.
Amount USD =
IF (
    HASONEVALUE ( Dim_Scenario[Scenario] ),
    SUM ( Fact_SC_PL[Amount USD] ),
    CALCULATE ( SUM ( Fact_SC_PL[Amount USD] ), Dim_Scenario[Scenario] = "Actual" )
)

Pairs =
IF (
    HASONEVALUE ( Dim_Scenario[Scenario] ),
    SUM ( Fact_SC_PL[Volume Pairs] ),
    CALCULATE ( SUM ( Fact_SC_PL[Volume Pairs] ), Dim_Scenario[Scenario] = "Actual" )
)

Display Divisor = SELECTEDVALUE ( Display_Unit[Divisor], 1 )

// Dynamic format string used by every USD amount measure (Format: Dynamic):
// SWITCH ( SELECTEDVALUE ( Display_Unit[Display Unit], "Unit" ),
//     "Thousand", "#,0.0,""K"";-#,0.0,""K""",
//     "Million", "#,0.00,""M"";-#,0.00,""M""",
//     "#,0;-#,0" )
```

```
// P&L lines - one measure per SMART account (codes from template 'Dashboard Guide')
Purchase Pairs = CALCULATE ( [Pairs], Dim_Account[Account Code] = "PURCH_PAIRS" )
Production Pairs = CALCULATE ( [Pairs], Dim_Account[Account Code] = "PROD_PAIRS" )
Total Pairs = [Purchase Pairs] + [Production Pairs] // "Purchase and Production Pairs"

Turnover SC = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_119001I" )
Turnover SC Markup = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_119002I" )
Net Turnover SC = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_119000T" )
COGS SC = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_219000T" )
Inventory Valuation = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_222000I" )
Factory Personnel = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_227100I" )
Other Factory Costs = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_227200I" )
Factory Costs = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_227100T" )
Recoveries of Factory = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_227600I" )
Factory Under Over Recovery = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_227000T" )
Freight Costs = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_225000I" )
PPV = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_224000I" )
Cost of Sales = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_200000T" )
Intercompany GP = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_IC0000T" )
Gross Profit = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_920000T" ) // Total Gross Profit
Marketing Expenses = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_310000T" )
Four Walls Expenses = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_4W0000T" )
Channel Expenses = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_CH0000T" )
Channel Operating Profit = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_CP0000T" )
Enabling Functions Expenses = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_OH0000T" )
Non Recurring Items = CALCULATE ( [Amount USD], Dim_Account[Account Code] = "PL_380000I" ) // mapping TBC (Q07)
```

5.3 DAX measures - profit levels, ratios, cost per pair, factory and sourcing

```
// Profit levels that the template marks as "Calculated"
EBITDA =
[Channel Operating Profit] + [Enabling Functions Expenses] // ASSUMPTION A07 (template sample = GP + Channel Expenses)

EBIT BNRI = [EBITDA] // template: EBIT BNRI = EBITDA (D&A line TBC, Q07)

EBIT = [EBIT BNRI] + [Non Recurring Items] // ASSUMPTION A08 - NRI signed as in SMART (template shows "minus")
```

```
// Financial KPIs (Section A of KPI file)
Turnover SC Markup % = DIVIDE ( [Turnover SC Markup], [Turnover SC] )
Gross Margin % = DIVIDE ( [Gross Profit], [Net Turnover SC] )
EBITDA Margin % = DIVIDE ( [EBITDA], [Net Turnover SC] )
EBIT BNRI Margin % = DIVIDE ( [EBIT BNRI], [Net Turnover SC] )
EBIT Margin % = DIVIDE ( [EBIT], [Net Turnover SC] )
Channel Operating Margin % = DIVIDE ( [Channel Operating Profit], [Net Turnover SC] )

// Cost Analytics - cost ratios shown as positive magnitudes (lower = better)
Cost of Sales % = DIVIDE ( - [Cost of Sales], [Net Turnover SC] ) // ASSUMPTION A06: "Turnover" = Net Turnover SC
Channel Cost % = DIVIDE ( - [Channel Expenses], [Net Turnover SC] ) // = template "Channel Expenses %" (A05)
Freight Cost per Pair = DIVIDE ( - [Freight Costs], [Total Pairs] ) // Freight / (Purchase + Production Pairs)
Factory Cost per Pair = DIVIDE ( - [Factory Costs], [Production Pairs] )
```

```
Recovery per Pair          = DIVIDE ( [Recoveries of Factory], [Production Pairs] )
Channel Expenses per Pair  = DIVIDE ( - [Channel Expenses], [Total Pairs] )
```

```
// Manufacturing KPIs - denominator = Turnover SC of the Manufacturing function (KPI file remark)
Factory Personnel Cost % =
CALCULATE (
    DIVIDE ( - [Factory Personnel], [Turnover SC] ),
    KEEPFILTERS ( Dim_SC_Function[SC Function] = "Manufacturing" )
)

Other Factory Cost % =
CALCULATE (
    DIVIDE ( - [Other Factory Costs], [Turnover SC] ),
    KEEPFILTERS ( Dim_SC_Function[SC Function] = "Manufacturing" )
)

// Procurement KPIs - denominator = Turnover SC of Imports + Local Sourcing (KPI file remark)
PPV % =
CALCULATE (
    DIVIDE ( [PPV], [Turnover SC] ),
    KEEPFILTERS ( Dim_SC_Function[SC Function] IN { "Imports", "Local Sourcing" } )
)

Import Mix % =
DIVIDE (
    CALCULATE ( [Purchase Pairs], KEEPFILTERS ( Dim_SC_Function[SC Function] = "Imports" ) ),
    CALCULATE ( [Purchase Pairs], KEEPFILTERS ( Dim_SC_Function[SC Function] IN { "Imports", "Local Sourcing" } ) )
)

Local Sourcing Mix % =
DIVIDE (
    CALCULATE ( [Purchase Pairs], KEEPFILTERS ( Dim_SC_Function[SC Function] = "Local Sourcing" ) ),
    CALCULATE ( [Purchase Pairs], KEEPFILTERS ( Dim_SC_Function[SC Function] IN { "Imports", "Local Sourcing" } ) )
)

Average Cost per Pair = DIVIDE ( - [COGS SC], [Total Pairs] )

Average Cost per Pair Import =
CALCULATE ( DIVIDE ( - [COGS SC], [Purchase Pairs] ), KEEPFILTERS ( Dim_SC_Function[SC Function] = "Imports" ) )

Average Cost per Pair LS =
CALCULATE ( DIVIDE ( - [COGS SC], [Purchase Pairs] ), KEEPFILTERS ( Dim_SC_Function[SC Function] = "Local Sourcing" ) )

Average Cost per Pair IHM =
CALCULATE ( DIVIDE ( - [COGS SC], [Production Pairs] ), KEEPFILTERS ( Dim_SC_Function[SC Function] = "Manufacturing" ) )
```

5.4 Variance vs Budget / LY, YTD - calculation groups and explicit equivalents

```
// Calculation group 'CG Scenario' (column [Comparison], precedence 10)
// Mirrors the template columns: Actual | Budget | LY | vs Bud % | vs LY %
// Item "Actual"
CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Actual" )
// Item "Budget"
CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Budget" )
// Item "LY" - SMART LY scenario if supplied, otherwise prior-year Actual (ASSUMPTION A03)
VAR _src = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "LY" )
RETURN
    IF (
        NOT ISBLANK ( _src ),
        _src,
        CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Actual", DATEADD ( Dim_Date[Date], -1, YEAR ) )
    )
// Item "Var vs Bud" (absolute; for % KPIs this is the difference in percentage points)
VAR _a = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Actual" )
VAR _b = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Budget" )
RETURN IF ( NOT ISBLANK ( _a ) && NOT ISBLANK ( _b ), _a - _b )
// Item "Var vs Bud %" (relative for amounts; pp for ratio KPIs)
VAR _a = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Actual" )
VAR _b = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Budget" )
VAR _isRatio = CONTAINSSTRING ( SELECTEDMEASUREFORMATSTRING (), "%" )
RETURN
    IF (
        ISBLANK ( _a ) || ISBLANK ( _b ), BLANK (), // shown as "n.a" (template)
        IF ( _isRatio, ( _a - _b ) * 100, DIVIDE ( _a - _b, ABS ( _b ) ) )
    )
// Format string expression:
// IF ( CONTAINSSTRING ( SELECTEDMEASUREFORMATSTRING (), "%" ), "+0.0 ""pp"";-0.0 ""pp""", "+0.0%;-0.0%" )
// Item "Var vs LY" / "Var vs LY %" - same pattern, with _b replaced by the LY logic:
VAR _a = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Actual" )
VAR _lySrc = CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "LY" )
VAR _ly =
    IF ( NOT ISBLANK ( _lySrc ), _lySrc,
        CALCULATE ( SELECTEDMEASURE (), Dim_Scenario[Scenario] = "Actual", DATEADD ( Dim_Date[Date], -1, YEAR ) ) )
```

```

VAR _isRatio = CONTAINSSTRING ( SELECTEDMEASUREFORMATSTRING (), "%" )
RETURN
    IF ( ISBLANK ( _a ) || ISBLANK ( _ly ), BLANK (),
        IF ( _isRatio, ( _a - _ly ) * 100, DIVIDE ( _a - _ly, ABS ( _ly ) ) ) )

// Calculation group 'CG Period' (column [Period], precedence 20 = applied outside CG Scenario)
// Mirrors template filter "Period: Monthly | Quarterly | YTD" (calendar year - ASSUMPTION A12)
// Item "Monthly"
SELECTEDMEASURE ()
// Item "Quarterly" (quarter-to-date)
CALCULATE ( SELECTEDMEASURE (), DATESQTD ( Dim_Date[Date] ) )
// Item "YTD"
CALCULATE ( SELECTEDMEASURE (), DATESYTD ( Dim_Date[Date] ) )

```

Explicit measures (same logic without calculation groups - used in the email query and tooltips)

```

// Explicit equivalents (used in email query / tooltips, and to show the logic without calc groups)
Net Turnover SC Budget = CALCULATE ( [Net Turnover SC], Dim_Scenario[Scenario] = "Budget" )
Net Turnover SC LY = CALCULATE ( [Net Turnover SC], 'CG Scenario'[Comparison] = "LY" )
Net Turnover SC vs Bud % = DIVIDE ( [Net Turnover SC] - [Net Turnover SC Budget], ABS ( [Net Turnover SC Budget] ) )
Net Turnover SC vs LY % = DIVIDE ( [Net Turnover SC] - [Net Turnover SC LY], ABS ( [Net Turnover SC LY] ) )
Net Turnover SC YTD = CALCULATE ( [Net Turnover SC], DATESYTD ( Dim_Date[Date] ) )
Net Turnover SC YTD Bud = CALCULATE ( [Net Turnover SC Budget], DATESYTD ( Dim_Date[Date] ) )
Net Turnover SC YTD LY = CALCULATE ( [Net Turnover SC], Dim_Scenario[Scenario] = "Actual",
    DATESYTD ( DATEADD ( Dim_Date[Date], -1, YEAR ) ) )
Freight Cost per Pair YTD = CALCULATE ( [Freight Cost per Pair], DATESYTD ( Dim_Date[Date] ) )
Gross Margin % vs Bud pp = ( [Gross Margin %] - CALCULATE ( [Gross Margin %], Dim_Scenario[Scenario] = "Budget" ) ) * 100

```

5.5 Exception management (traffic lights), P&L matrix and KPI selector

```

// Exception management - traffic lights (KPI file section C). Comparison bases = ASSUMPTION A11.
Alert GP Margin =
VAR _a = [Gross Margin %]
VAR _b = CALCULATE ( [Gross Margin %], 'CG Scenario'[Comparison] = "Budget" )
VAR _ly = CALCULATE ( [Gross Margin %], 'CG Scenario'[Comparison] = "LY" )
RETURN IF ( ( NOT ISBLANK ( _b ) && _a - _b < -0.01 ) || ( NOT ISBLANK ( _ly ) && _a - _ly < -0.01 ), "Red", "Green" )

Alert EBITDA = IF ( [EBITDA] < 0, "Red", "Green" )

Alert Freight =
VAR _a = [Freight Cost per Pair]
VAR _ly = CALCULATE ( [Freight Cost per Pair], 'CG Scenario'[Comparison] = "LY" )
RETURN IF ( DIVIDE ( _a - _ly, _ly ) > 0.10, "Amber", "Green" )

Alert PPV =
VAR _a = [PPV]
VAR _b = CALCULATE ( [PPV], Dim_Scenario[Scenario] = "Budget" )
RETURN IF ( DIVIDE ( _a - _b, ABS ( _b ) ) < -0.05, "Amber", "Green" ) // PPV positive = favourable

Alert Recovery =
VAR _a = [Recoveries of Factory]
VAR _b = CALCULATE ( [Recoveries of Factory], Dim_Scenario[Scenario] = "Budget" )
RETURN IF ( ABS ( DIVIDE ( _a - _b, _b ) ) > 0.10, "Red", "Green" )

Alert Colour (hex) = // used by conditional formatting "Field value"
SWITCH ( [Alert GP Margin], "Red", "#C62828", "Amber", "#E0A100", "#1E8E3E" )

OpCos Flagged EBITDA = COUNTROWS ( FILTER ( VALUES ( Dim_OpCo[OpCo Code] ), [Alert EBITDA] = "Red" ) )
OpCos Flagged PPV = COUNTROWS ( FILTER ( VALUES ( Dim_OpCo[OpCo Code] ), [Alert PPV] = "Amber" ) )

```

```

// Detailed P&L matrix: rows = PL_Layout[Row Label] (sorted by PL_Layout[Sort]), columns = 'CG Scenario'[Comparison]
P&L Row Value =
VAR _type = SELECTEDVALUE ( PL_Layout[Row Type] )
VAR _key = SELECTEDVALUE ( PL_Layout[Account Code / Measure] )
RETURN
    SWITCH (
        TRUE (),
        _type = "Account" && _key IN { "PURCH_PAIRS", "PROD_PAIRS" }, CALCULATE ( [Pairs], Dim_Account[Account Code] =
            _key ),
        _type = "Account", CALCULATE ( [Amount USD], Dim_Account[Account Code] = _key ),
        _key = "Total Pairs", [Total Pairs],
        _key = "Gross Margin %", [Gross Margin %],
        _key = "Channel Cost %", [Channel Cost %],
        _key = "Channel Operating Margin %", [Channel Operating Margin %],
        _key = "EBITDA", [EBITDA],
        _key = "EBITDA Margin %", [EBITDA Margin %],
        _key = "EBIT BNRI", [EBIT BNRI],
        _key = "EBIT BNRI Margin %", [EBIT BNRI Margin %],
        _key = "EBIT", [EBIT],
        _key = "EBIT Margin %", [EBIT Margin %]
    )
// Dynamic format string: SWITCH ( SELECTEDVALUE ( PL_Layout[Format] ), "%", "0.0%", "#,0", "#,0", <USD dynamic format> )

```



```
// Trend page: KPI slicer bound to Dim_KPI (single select)
KPI Value =
SWITCH (
    SELECTEDVALUE ( Dim_KPI[KPI ID] ),
    "K01", [Net Turnover SC],          "K02", [Turnover SC Markup %],      "K03", [Gross Profit],
    "K04", [Gross Margin %],          "K05", [EBITDA],                    "K06", [EBITDA Margin %],
    "K07", [EBIT],                    "K08", [EBIT Margin %],            "K09", [Cost of Sales],
    "K10", [Purchase Pairs],          "K11", [Production Pairs],         "K12", [Total Pairs],
    "K13", [Freight Cost per Pair],    "K14", [Factory Cost per Pair],     "K15", [Recovery per Pair],
    "K16", [Channel Expenses per Pair], "K17", [Channel Cost %],            "K18", [Cost of Sales %],
    "K19", [Factory Under Over Recovery], "K20", [Factory Personnel Cost %], "K21", [Other Factory Cost %],
    "K22", [PPV %],                    "K23", [Import Mix %],             "K24", [Local Sourcing Mix %],
    "K25", [Average Cost per Pair],    "K26", [Average Cost per Pair Import],
    "K27", [Average Cost per Pair LS], "K28", [Average Cost per Pair IHM]
)
KPI Definition (tooltip) = SELECTEDVALUE ( Dim_KPI[Description (source)] ) & " | " & SELECTEDVALUE ( Dim_KPI[Formula] )
```

Card measures. Each Summary card needs six values (Monthly / YTD x Actual / vs Bud % / vs LY %). These are generated automatically by the Tabular Editor script below from the annotation SummaryCard = Y, so adding or swapping a card KPI needs no manual DAX.

```
// Tabular Editor (C#) script - generates the 6 card measures (Monthly/YTD x Actual/vs Bud %/vs LY %)
// for every measure annotated SummaryCard = "Y" - no manual measure writing when a card KPI is added.
/*
foreach (var m in Model.AllMeasures.Where(x => x.GetAnnotation("SummaryCard") == "Y").ToList())
{
    foreach (var p in new[] { "Monthly", "YTD" })
    foreach (var s in new[] { "Actual", "Var vs Bud %", "Var vs LY %" })
    {
        var n = m.Name + " | " + p + " | " + s;
        var expr = "CALCULATE ( " + m.DaxObjectFullName + ", 'CG Scenario'[Comparison] = \"\" + s
                    + "\", 'CG Period'[Period] = \"\" + p + \"\" )";
        var nm = m.Table.AddMeasure(n, expr, "Summary Cards");
        nm.FormatString = s == "Actual" ? m.FormatString
                               : (m.FormatString.Contains("%") ? "+0.0 \"pp\";-0.0 \"pp\" : "+0.0%;-0.0%");
    }
}
*/
```

6. Power Query - Ingestion, Validation and Transformation

Power Query runs in a **Power BI Dataflow** ('SC_PL_Staging'). The semantic model reads the dataflow tables. For the prototype, the same M runs directly in Power BI Desktop against a local folder (Folder.Files) holding the sample CSV.

Step	Query	What it does	DQ rules
1	Parameters	SharePoint site, library path, list of mandatory columns	-
2	Stg_Files	Lists files in 01_Inbox and 03_Archive; parses period and version from SCPL_YYYYMM_vN; keeps the latest version per month; reads each file; lists missing columns	DQ02, DQ03
3	Stg_Extract	Expands all valid files; trims and uppercases codes; null-safe typing with try ... otherwise; adds Row No for traceability	DQ09
4	DQ_Exceptions	Row-level rules (mandatory fields, period, scenario, currency, numeric, pairs, master-data lookups), duplicate keys, file-level errors, missing active OpCos	DQ04-DQ11
5	Fact_SC_PL	Removes rows with Critical issues; keeps only Published periods; builds Date Key; splits Amount into Amount USD and Volume Pairs; maps function codes	Gate
6	Dimensions / lists	Dim_Date generated; Dim_OpCo, Dim_Account, Dim_SC_Function, User_Access, Publish_Control from SharePoint lists	DQ06
7	Refresh_Stamp	UTC time stamp of every refresh for 'Last Refresh Time'	DQ14

6.1 Parameters and functions

```
// Parameters (Manage Parameters)
SharePointSite = "https://<tenant>.sharepoint.com/sites/SC-Finance" meta [IsParameterQuery = true, Type = "Text",
IsParameterQueryRequired = true]
LibraryPath     = "/SC_PL/" meta [IsParameterQuery = true, Type = "Text",
IsParameterQueryRequired = true]

// Mandatory columns - aligned to the template (Accounts | Channel | Accdetail) + filters (Year, Month, Region, Country,
BU)
RequiredColumns = {"Year", "Month", "Region", "Country", "OpCo Code", "BU", "SC Function",
```

```

"Account", "Channel", "AccDetail", "Scenario", "Currency", "Amount"}

// fxClean - null-safe trim
fxClean = (v as any) as nullable text => if v = null then null else Text.Trim(Text.From(v))

// fxReadExtract - reads one .xlsx (first sheet) or .csv file and promotes headers
fxReadExtract = (content as binary, fileName as text) as table =>
let
    isCsv = Text.EndsWith(Text.Lower(fileName), ".csv"),
    raw = if isCsv
        then Csv.Document(content, [Delimiter = ",", Encoding = 65001, QuoteStyle = QuoteStyle.Csv])
        else Table.SelectRows(Excel.Workbook(content, null, true), each [Kind] = "Sheet"){0}[Data],
    promoted = Table.PromoteHeaders(raw, [PromoteAllScalars = true]),
    names = Table.TransformColumnNames(promoted, Text.Trim)
in
    names

// fxMissingColumns - list of mandatory columns not present in a file (DQ03)
fxMissingColumns = (t as table) as list => List.Difference(RequiredColumns, Table.ColumnNames(t))

```

6.2 Stg_Files - file discovery, versioning, structure check

```

// Stg_Files - one row per candidate file; latest version per period; file-level checks (DQ02, DQ03)
let
    Source = SharePoint.Files(SharePointSite, [ApiVersion = 15]),
    InScope = Table.SelectRows(Source, each
        Text.Contains([Folder Path], LibraryPath & "01_Inbox/")
        or Text.Contains([Folder Path], LibraryPath & "03_Archive/")),
    AddPeriod = Table.AddColumn(InScope, "File Period",
        each try Int64.From(Text.Middle([Name], 5, 6)) otherwise null, Int64.Type),
    AddVersion = Table.AddColumn(AddPeriod, "File Version",
        each try Int64.From(Text.BetweenDelimiters([Name], "_v", ".")) otherwise 1, Int64.Type),
    AddNameOK = Table.AddColumn(AddVersion, "Name Valid", each
        Text.StartsWith([Name], "SCPL_")
        and [File Period] <> null
        and List.Contains({"xlsx", ".csv"}, Text.Lower([Extension])), type logical),
    ValidNames = Table.SelectRows(AddNameOK, each [Name Valid]),
    // keep only the highest version per reporting month (v2 replaces v1 - auditability kept in Archive)
    Grouped = Table.Group(ValidNames, {"File Period"},
        {"Latest", each Table.First(Table.Sort(_, {"File Version", Order.Descending})), type record}),
    Latest = Table.FromRecords(Grouped[Latest]),
    AddData = Table.AddColumn(Latest, "Data", each try fxReadExtract([Content], [Name]) otherwise null),
    AddMissing = Table.AddColumn(AddData, "Missing Columns", each
        if [Data] = null then "UNREADABLE FILE" else Text.Combine(fxMissingColumns([Data]), ", "), type
text),
    Result = Table.SelectColumns(AddMissing, {"Name", "Folder Path", "File Period", "File Version", "Date modified",
    "Data", "Missing Columns"})
in
    Result

```

6.3 Stg_Extract - combine, cleanse, type

```

// Stg_Extract - combined, cleansed, typed rows from all structurally valid files
let
    Files = Table.SelectRows(Stg_Files, each [Missing Columns] = ""),
    Keep = Table.SelectColumns(Files, {"Name", "File Period", "File Version", "Data"}),
    Expanded = Table.ExpandTableColumn(Keep, "Data", RequiredColumns),
    Cleaned = Table.TransformColumns(Expanded, {
        {"Region", fxClean, type nullable text}, {"Country", fxClean, type nullable text},
        {"OpCo Code", each Text.Upper(fxClean(_)) ?? "", type text},
        {"BU", fxClean, type nullable text}, {"SC Function", fxClean, type nullable text},
        {"Account", each Text.Upper(fxClean(_)) ?? "", type text},
        {"Channel", fxClean, type nullable text}, {"AccDetail", fxClean, type nullable text},
        {"Scenario", fxClean, type nullable text}, {"Currency", each Text.Upper(fxClean(_)) ?? "", type
text}},
    Typed = Table.TransformColumns(Cleaned, {
        {"Year", each try Int64.From(_) otherwise null, type nullable number},
        {"Month", each try Int64.From(_) otherwise null, type nullable number},
        {"Amount", each try Number.From(_, "en-US") otherwise null, type nullable number}},
    WithRowNo = Table.AddIndexColumn(Typed, "Row No", 1, 1, Int64.Type)
in
    WithRowNo

```

6.4 DQ_Exceptions - validation rules

```

// DQ_Exceptions - every validation failure, one row per issue (feeds DQ page, approval flow, and fact filter)
let
    S = Stg_Extract,
    OpCoList = List.Buffer(Dim_OpCo[OpCo Code]),
    AccountList = List.Buffer(Dim_Account[Account Code]),
    FunctionList = List.Buffer(Dim_SC_Function[SC Function]),
    Mandatory = {"Year", "Month", "Country", "OpCo Code", "SC Function", "Account", "Scenario", "Currency",
    "Amount"},
    // ---- row-level rules

```

```

AddIssues = Table.AddColumn(S, "Issues", each
    List.RemoveNulls({
        if List.AnyTrue(List.Transform(Mandatory, (c) => Record.Field(_, c) = null or Record.Field(_, c) = ""))
            then "DQ04|Critical|Mandatory field blank (OpCo / Country / Month / other)" else null,
        if [Year] = null or [Month] = null or [Month] < 1 or [Month] > 12
            then "DQ05|Critical|Invalid year / month" else null,
        if [Year] <> null and [Month] <> null and [Year] * 100 + [Month] <> [File Period]
            then "DQ05|Critical|Row period differs from file period" else null,
        if not List.Contains({"Actual", "Budget", "LY"}, [Scenario])
            then "DQ07|Critical|Invalid scenario '" & [Scenario] ?? "" & "' else null,
        if [Currency] <> "USD"
            then "DQ08|Critical|Currency '" & [Currency] & "' is not USD" else null,
        if [Amount] = null
            then "DQ09|Critical|Amount is not numeric" else null,
        if List.Contains({"PURCH_PAIRS", "PROD_PAIRS"}, [Account]) and [Amount] <> null
            and ([Amount] < 0 or Number.Round([Amount]) <> [Amount])
            then "DQ09|Critical|Pairs must be non-negative whole numbers" else null,
        if [OpCo Code] <> "" and not List.Contains(OpCoList, [OpCo Code])
            then "DQ06|High|OpCo not in Dim_OpCo (new OpCo? map it)" else null,
        if [Account] <> "" and not List.Contains(AccountList, [Account])
            then "DQ06|High|Account not in Dim_Account (map it)" else null,
        if [SC Function] <> null and not List.Contains(FunctionList, [SC Function])
            then "DQ06|High|SC Function not in Dim_SC_Function" else null
    })),
RowIssues = Table.ExpandListColumn(Table.SelectRows(AddIssues, each List.Count([Issues]) > 0), "Issues"),
RowSplit = Table.SplitColumn(RowIssues, "Issues", Splitter.SplitTextByDelimiter("|"), {"Rule ID", "Severity",
"Message"}),
RowOut = Table.SelectColumns(RowSplit, {"Name", "File Period", "Row No", "OpCo Code", "Account", "Scenario", "Rule
ID", "Severity", "Message"}),
// ---- duplicates (DQ10)
KeyCols = {"File Period", "OpCo Code", "SC Function", "Account", "Scenario", "Channel", "AccDetail"},
Dup = Table.Group(S, KeyCols, {"Count", Table.RowCount, Int64.Type}, {"Name", each List.First([Name]), type
text}),
DupOut = Table.AddColumn(
    Table.AddColumn(
        Table.AddColumn(Table.SelectRows(Dup, each [Count] > 1), "Rule ID", each "DQ10"),
        "Severity", each "High"),
    "Message", each "Duplicate key - " & Text.From([Count]) & " rows"),
// ---- file-level (DQ02 name/format handled by flow; DQ03 missing columns / unreadable here)
FileOut = Table.AddColumn(
    Table.AddColumn(
        Table.AddColumn(Table.SelectRows(Stg_Files, each [Missing Columns] <> ""), "Rule ID", each
"DQ03"),
        "Severity", each "Critical"),
    "Message", each "Missing columns: " & [Missing Columns]),
// ---- completeness (DQ11): active OpCos with no Actual rows for the newest file period
NewestPeriod = List.Max(S[File Period]),
LoadedOpCos = List.Distinct(Table.SelectRows(S, each [File Period] = NewestPeriod and [Scenario] = "Actual")[OpCo
Code]),
MissingOpCos = Table.SelectRows(Dim_OpCo, each [Active] = "Y" and not List.Contains(LoadedOpCos, [OpCo Code])),
MissOut = Table.AddColumn(
    Table.AddColumn(
        Table.AddColumn(
            Table.AddColumn(Table.SelectColumns(MissingOpCos, {"OpCo Code"}), "File Period", each
NewestPeriod),
                "Rule ID", each "DQ11"),
            "Severity", each "High"),
    "Message", each "Active OpCo has no Actual rows in file"),
Combined = Table.Combine({RowOut, DupOut, FileOut, MissOut}),
WithStatus = Table.AddColumn(Combined, "Status", each "Open", type text),
Typed = Table.TransformColumnTypes(WithStatus, {"Rule ID", type text}, {"Severity", type text}, {"Message", type
text}),
in
Typed

```

6.5 Fact_SC_PL - publish gate and transformation

```

// Fact_SC_PL - only rows without Critical issues, only periods the Administrator has Published
let
    S = Stg_Extract,
    CriticalRows = Table.SelectRows(DQ_Exceptions, each [Severity] = "Critical" and [Row No] <> null),
    NoCritical = Table.NestedJoin(S, {"Name", "Row No"}, CriticalRows, {"Name", "Row No"}, "Err", JoinKind.LeftAnti),
    PublishedPeriods = List.Buffer(Table.SelectRows(Publish_Control, each [Status] = "Published")[Period]),
    Published = Table.SelectRows(NoCritical, each List.Contains(PublishedPeriods, [File Period])),
    AddDate = Table.AddColumn(Published, "Date Key", each #date([Year], [Month], 1), type date),
    JoinAcc = Table.NestedJoin(AddDate, {"Account"}, Dim_Account, {"Account Code"}, "Acc", JoinKind.Inner),
    ExpAcc = Table.ExpandTableColumn(JoinAcc, "Acc", {"Measure Type"}),
    JoinFunc = Table.NestedJoin(ExpAcc, {"SC Function"}, Dim_SC_Function, {"SC Function"}, "Fn", JoinKind.Inner),
    ExpFunc = Table.ExpandTableColumn(JoinFunc, "Fn", {"Function Code"}),
    // split value column into USD amount and pairs (template rows PURCH_PAIRS / PROD_PAIRS)
    AddUSD = Table.AddColumn(ExpFunc, "Amount USD", each if [Measure Type] = "USD" then [Amount] else null, type
nullable number),
    AddPairs = Table.AddColumn(AddUSD, "Volume Pairs", each if [Measure Type] = "Pairs" then [Amount] else null, type
nullable number),
    Final = Table.SelectColumns(AddPairs, {"Date Key", "OpCo Code", "Function Code", "Account", "Scenario",
"Channel", "AccDetail", "Amount USD", "Volume Pairs", "Name", "File
Version"}),

```

```

Renamed      = Table.RenameColumns(Final, {"Account", "Account Code"}, {"Function Code", "SC Function Code"}, {"Name",
"Source File"})
in
Renamed

```

6.6 Dim_Date and Refresh_Stamp

```

// Dim_Date - daily calendar (mark as date table); fact joins on first day of month
let
    StartDate = #date(2024, 1, 1),
    EndDate   = Date.EndOfYear(Date.AddYears(DateTime.Date(DateTime.LocalNow()), 1)),
    Dates     = List.Dates(StartDate, Duration.Days(EndDate - StartDate) + 1, #duration(1, 0, 0, 0)),
    T         = Table.FromList(Dates, Splitter.SplitByNothing(), {"Date"}),
    Typed     = Table.TransformColumnTypes(T, {"Date", type date}),
    AddYear   = Table.AddColumn(Typed, "Year", each Date.Year([Date]), Int64.Type),
    AddMonth  = Table.AddColumn(AddYear, "Month", each Date.Month([Date]), Int64.Type),
    AddMName  = Table.AddColumn(AddMonth, "Month Name", each Date.MonthName([Date], "en-US"), type text),
    AddMShort = Table.AddColumn(AddMName, "Month Short", each Text.Start(Date.MonthName([Date], "en-US"), 3), type text),
    AddQtr    = Table.AddColumn(AddMShort, "Quarter", each "Q" & Text.From(Date.QuarterOfYear([Date])), type text),
    AddPeriod = Table.AddColumn(AddQtr, "Period", each Date.Year([Date]) * 100 + Date.Month([Date]), Int64.Type)
in
    AddPeriod

// Refresh_Stamp - "Last Refresh Time" shown on every page (as in template)
let
    Source = #table(type table [{"Refreshed At (UTC)" = datetimezone}], {[DateTimeZone.UtcNow()]})
in
    Source

```

User_Access and Publish_Control are read with SharePoint.Tables. Their M code is in the companion file SC_PL_PowerQuery_M.pq.

7. Dashboard Design

7.1 Report structure and global standards

#	Page	Mirrors (template / KPI file)	Audience
0	Home & Dashboard Guide	Template 'Dashboard Guide' (navigation, general notes, KPI glossary from Dim_KPI)	All
1	Executive Summary	Template 'Summary' tab - Sections A-D, 12 KPI cards	All
2	Detailed P&L	Template 'Detailed - Supply Chain' (+ Imports / LS / Manufacturing / Global Sourcing buttons)	All
3	OpCo / Country Performance	KPI file Section C (Exception Management) + Section A by OpCo	All
4	IHM / Manufacturing	KPI file 'Manufacturing KPIs' + factory Cost Analytics KPIs	All (RLS: Manufacturing)
5	Sourcing & Procurement	KPI file 'Procurement KPIs' (PPV %, Import / Local mix, Average cost per pair)	All (RLS: Imports / LS / GS)
6	Trend Analysis (Monthly / YTD)	Template 'Trend Analysis' tab (decomposition tree, Actual vs Budget trend, cost treemap, cost trend)	All
7	Data Quality & Refresh Status	Brief requirement (Admin); Deliverable 11	Admin audience only
T	Tooltip: KPI definition	Template '(i)' icon - definition, formula, 13-month sparkline	All

- **Canvas** 16:9 (1280 x 720) for desktop and web; a dedicated **mobile layout** for Summary and OpCo pages. The theme (SC_PL_PowerBI_Theme.json) reproduces the template teal (#008F94 / #DDF3F1).
- **Header band on every page:** page title in template wording ('Supply Chain P&L | Summary'), subtitle '<Region/Country/OpCo selection> - <Month Year> | USD', navigation buttons, Unit / Thousand / Million toggle (Display_Unit slicer styled as buttons), (i) info icon, reset-filters button.
- **Left filter pane** (synchronised across pages) as in the template: Time Filter, Location Filter, Other Filter, Period and Last Refresh Time.
- **Colour rules:** favourable = green, adverse = red, warning = amber (polarity-aware). Actual = teal / dark red line, Budget = grey, LY = light teal, as in the template trend image.
- **USD labelling:** every financial card title carries a unit label (USD M / USD K / USD per pair). Axis titles and matrix headers state USD.

7.2 Page 1 - Executive Summary (mirrors the 'Summary' tab)

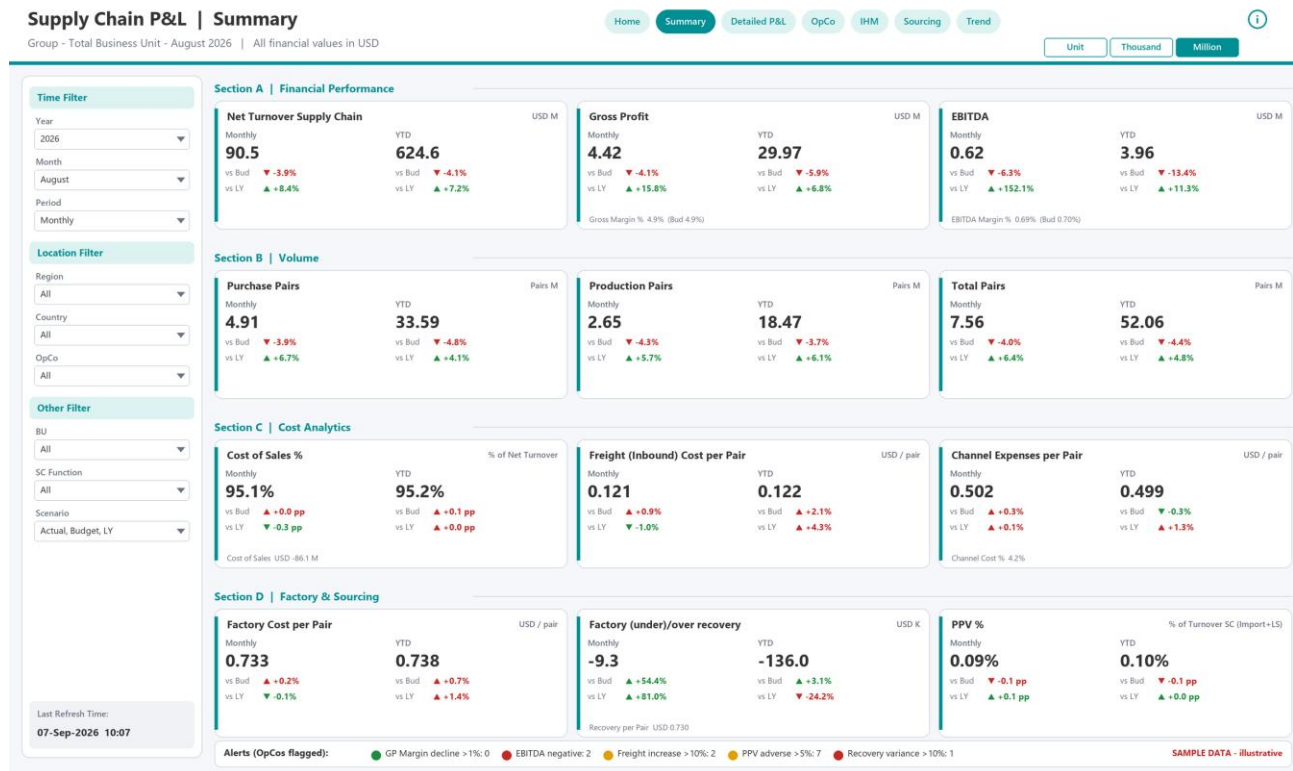


Figure 3 - Executive Summary page mock-up (SAMPLE DATA), layout taken from the template Summary tab

Section (template)	Card 1	Card 2	Card 3
A - Financial Performance	Net Turnover Supply Chain (USD)	Gross Profit (USD) + Gross Margin % sub-line	EBITDA (USD) + EBITDA Margin % sub-line
B - Volume	Purchase Pairs	Production Pairs	Total Pairs
C - Cost Analytics	Cost of Sales % (+ Cost of Sales USD)	Freight (Inbound) Cost per Pair (USD)	Channel Expenses per Pair (USD) + Channel Cost %
D - Factory & Sourcing	Factory Cost per Pair (USD)	Factory (under)/over recovery (USD) + Recovery per Pair	PPV %

- Card anatomy** (as in the template): KPI name and unit | Monthly value, vs Bud, vs LY | YTD value, vs Bud, vs LY, with arrow and colour. Built with the Card (new) visual and the generated card measures.
- Alert strip:** the five traffic-light rules with the number of OpCos flagged. Clicking a rule opens OpCo / Country Performance filtered to the flagged OpCos (drill-through).
- Interactions:** tooltip page with the KPI definition (from Dim_KPI) and a 13-month sparkline; drill-through from any card to Detailed P&L or Trend with the filters kept.
- Sample reading (August 2026, SAMPLE DATA): Net Turnover SC USD 90.5 M, -3.9% vs Budget and +8.4% vs LY; Gross Margin 4.9%.

7.3 Pages 2 to 6 - layout, visuals and interactions

Page	Layout and visuals	Filters / interactions
2 Detailed P&L	Button bar 'Supply Chain Imports Local Sourcing Manufacturing Global Sourcing' (SC Function slicer as buttons; Supply Chain = all). Matrix: rows = PL_Layout (template order, subtotal rows in bold, % rows in italics); columns = CG Scenario (Actual, Budget, LY, vs Bud %, vs LY %). Icon conditional formatting on the variance columns; 'n.a.' for blank bases.	Period toggle Monthly / Quarterly / YTD (CG Period). Expand columns to Region > Country > OpCo. Drill-through target from every page. Export to Excel (summarised data, RLS applied).
3 OpCo / Country Performance	Matrix OpCo x KPIs (Net Turnover, GP, GM %, EBITDA, Freight / pair, Factory cost / pair, PPV %) with traffic-light icons (Alert measures). Bar chart: GM % vs Budget (pp) by OpCo, sorted. Waterfall: Gross Profit variance vs Budget by Region > OpCo. Filled map by Country (optional).	Region > Country > OpCo drill-down; multi-select OpCos; cross-highlighting; right-click drill-through to Detailed P&L.

Page	Layout and visuals	Filters / interactions
4 IHM / Manufacturing	Cards: Production Pairs, Factory Cost per Pair, Recovery per Pair, Factory (under)/over recovery, Factory Personnel Cost %, Other Factory Cost %, Average Cost per Pair - IHM. Waterfall: Factory Personnel + Other Factory Costs -> Recoveries -> (under)/over recovery. Combo chart: monthly Factory Cost per Pair vs Recovery per Pair (Actual line, Budget dashed). Matrix by factory OpCo.	Page filter SC Function = Manufacturing. Recovery variance alert highlighted. Tooltips with definitions.
5 Sourcing & Procurement	Cards: Purchase Pairs, PPV %, Import Mix %, Local Sourcing Mix %, Average Cost per Pair (Import, LS). 100% stacked column: Import vs LS purchase pairs by OpCo. Line: PPV % by month (Actual / Budget / LY). Scatter: Average cost per pair vs purchase pairs by OpCo (benchmarking).	Page filter SC Function in {Imports, Local Sourcing, Global Sourcing}. PPV adverse alert highlighted.
6 Trend Analysis	As in the template: decomposition tree (KPI Value by Region -> OpCo -> Month); line chart Actual vs Budget vs LY by month (KPI selector from Dim_KPI) with anomaly detection; Cost Breakdown treemap (Freight, Factory Personnel, Factory OH, Channel Expenses, PPV); Cost Trend Analysis lines (Freight, PPV, Factory Cost).	KPI single-select slicer; Month 'All' (template); Monthly / YTD switch; drill Year > Quarter > Month; AI decomposition tree.

7.4 Filters

Filter	Field	Type	Multi-select	Pages
Year	Dim_Date[Year]	Dropdown	No (single)	All
Month	Dim_Date[Period Selector] (defaults to 'Latest Month')	Dropdown	No; Trend allows All	All
Period	CG Period[Period] - Monthly / Quarterly / YTD	Buttons	No	Detailed, OpCo, IHM, Sourcing, Trend
Region	Dim_OpCo[Region]	Dropdown	Yes	All
Country	Dim_OpCo[Country]	Dropdown + search	Yes	All
OpCo	Dim_OpCo[OpCo Name]	Dropdown + search	Yes	All
BU	Dim_OpCo[BU] - XYZ Ltd / AW Lab	Dropdown	Yes	All
SC Function	Dim_SC_Function[SC Function]	Buttons / dropdown	Yes	All
KPI	Dim_KPI[KPI] (grouped by KPI-file section)	Dropdown	No (Trend); Yes (OpCo matrix)	Trend, OpCo
Scenario	CG Scenario[Comparison]	Buttons	Yes	Detailed, Trend
Display unit	Display_Unit[Display Unit] - Unit / Thousand / Million	Buttons	No	All

All slicers are synchronised across pages (Sync slicers). Viewers can save the current filter state as a **personal bookmark** in the app. A 'Reset' button restores the default view (latest published month, all permitted OpCos).

8. User Access Model

8.1 Roles

Role	Who	Power BI implementation
Administrator	Global SC Finance Controller (plus one named deputy)	Workspace Admin; model role 'SC Admin' (no filter); Admin app audience (includes DQ page); owner of the SharePoint lists and flows
Viewer	Country SC Head, IHM Head, Logistics Head, Quality Head, Sourcing Head, Finance Head	No workspace role; member of Entra group SG-SCPL-Viewers (Viewer app audience); model role 'SC Viewer' (dynamic RLS); no Build permission
Platform owner (support)	BI CoE / IT	Tenant settings, gateway (if needed), capacity, deployment pipeline (Dev > Test > Prod)

8.2 Permissions matrix

Capability	Administrator	Viewer
------------	---------------	--------

Capability	Administrator	Viewer
View report pages, filter, multi-select, drill down / through, tooltips	Yes	Yes
View Monthly / QTD / YTD, Actual / Budget / LY	Yes	Yes
Personal bookmarks / personalised views	Yes	Yes
Export permitted data (summarised, RLS applied)	Yes	Yes (summarised only)
Subscribe to own email / alerts	Yes	Yes
See all OpCos / Countries / Functions	Yes	No - only assigned scope
Data Quality & Refresh Status page	Yes	No
Approve / reject monthly publication	Yes	No
Create, activate / deactivate, delete users; assign role and scope	Yes	No
Maintain master data (OpCo, account mapping, KPI catalogue)	Yes	No
Change data model, KPI definitions, global calculations	Via controlled release (Dev > Test > Prod)	No
Change source data, security, global report design	Via controlled release	No

8.3 Dynamic Row-Level Security

Security comes from the **User_Access** table, a SharePoint List maintained by the Administrator. No user, country or OpCo is hard-coded. 'ALL' is a wildcard. A user may have several rows (scopes). Access is removed by setting Active = N or Valid To in the past, without deleting history.

User Email (SAMPLE)	Role title	App role	Region	OpCo	SC Function	Active
global.controller@company.example	Global SC Finance Controller	Administrator	ALL	ALL	ALL	Y
bd.schead@company.example	Country SC Head - Bangladesh	Viewer	ALL	BD01	ALL	Y
apac.ihmhead@company.example	IHM Head - APAC	Viewer	Asia-Pacific	ALL	Manufacturing	Y
logistics.head@company.example	Logistics Head - Global	Viewer	ALL	ALL	ALL	Y
quality.head@company.example	Quality Head - Global	Viewer	ALL	ALL	Manufacturing	Y
sourcing.head@company.example	Sourcing Head - Global	Viewer	ALL	ALL	Imports	Y
sourcing.head@company.example	Sourcing Head - Global	Viewer	ALL	ALL	Local Sourcing	Y
sourcing.head@company.example	Sourcing Head - Global	Viewer	ALL	ALL	Global Sourcing	Y
latam.finance@company.example	Finance Head - Latam	Viewer	Latam	ALL	ALL	Y
in.schead@company.example	Country SC Head - India	Viewer	ALL	IN01	ALL	N

RLS role definitions (Manage roles)

```
// Role "SC Viewer" - table filter on Dim_OpCo
VAR _u = USERPRINCIPALNAME ()
RETURN
    NOT ISEMPTY (
        FILTER (
            User_Access,
            User_Access[User Email] = _u                                // DAX "=" is case-insensitive
            && User_Access[Active] = "Y"
            && TODAY () >= User_Access[Valid From]
            && TODAY () <= User_Access[Valid To]
            && ( User_Access[Region] = "ALL" || User_Access[Region] = Dim_OpCo[Region] )
            && ( User_Access[OpCo Code] = "ALL" || User_Access[OpCo Code] = Dim_OpCo[OpCo Code] )
        )
    )

// Role "SC Viewer" - table filter on Dim_SC_Function
VAR _u = USERPRINCIPALNAME ()
RETURN
    NOT ISEMPTY (
        FILTER (
            User_Access,
            User_Access[User Email] = _u
            && User_Access[Active] = "Y"
            && TODAY () >= User_Access[Valid From]
            && TODAY () <= User_Access[Valid To]
            && ( User_Access[SC Function] = "ALL" || User_Access[SC Function] = Dim_SC_Function[SC Function] )
        )
    )

// Role "SC Viewer" - table filter on User_Access (users can only see their own mapping rows)
```

```

User_Access[User Email] = USERPRINCIPALNAME ()

// Role "SC Viewer" - Admin tables hidden from viewers
// Publish_Control, Refresh_Log, DQ_Exceptions : FALSE ()

// Role "SC Admin" - no filter (Global SC Finance Controller). Workspace Admin/Member are not subject to RLS anyway.
TRUE ()

// Optional stricter variant (exact OpCo x Function pairs) - filter on Fact_SC_PL instead of the two dimension filters
VAR _u = USERPRINCIPALNAME ()
VAR _region = RELATED ( Dim_OpCo[Region] )
VAR _opco = Fact_SC_PL[OpCo Code]
VAR _func = RELATED ( Dim_SC_Function[SC Function] )
RETURN
    NOT ISEMPTY (
        FILTER (
            User_Access,
            User_Access[User Email] = _u && User_Access[Active] = "Y"
            && ( User_Access[Region] = "ALL" || User_Access[Region] = _region )
            && ( User_Access[OpCo Code] = "ALL" || User_Access[OpCo Code] = _opco )
            && ( User_Access[SC Function] = "ALL" || User_Access[SC Function] = _func )
        )
    )

```

- The filters on Dim_OpCo and Dim_SC_Function propagate one-to-many to Fact_SC_PL, so every visual, export and email query respects the user's scope.
- The combination is OpCo-set x Function-set. Where a user needs exact pairs (e.g. Bangladesh all functions, but only IHM in Pakistan), the optional fact-level variant above is used.
- Testing: 'View as' > 'Other user' for each sample user, before every release. The Administrator checks access with the permission matrix above.
- Viewers are not given a workspace role (workspace roles bypass RLS) and do not get Build permission, so they cannot change the model or create reports on it.

8.4 User lifecycle (Administrator)

Action	How	Automation
Create user	New item in SCPL_User_Access (email, role, Region / OpCo / Function, Valid From)	Flow SCPL-04 adds the user to SG-SCPL-Viewers and sends a welcome email with the app link
Activate / deactivate	Set Active = Y / N (or Valid To)	SCPL-04 adds / removes group membership; RLS denies access immediately after the next refresh
Change scope	Edit Region / OpCo / Function or add a row	Picked up at the next refresh (a manual refresh can be triggered from the list)
Delete user	Delete the item (kept in list version history for audit)	SCPL-04 removes group membership
Quarterly access review	Admin reviews the list against the HR leavers report	Reminder flow each quarter

9. Monthly Operating Process

9.1 Monthly calendar (working days after month end)

When	Activity	Tool	Responsible
WD1-WD2	OpCo month-end close; SC P&L lines posted in SMART in USD	SMART	OpCo Finance
WD3 08:00	SMART scheduled extract to SharePoint 01_Inbox (SCPL_YYYYMM_v1)	SMART scheduler	SMART / IT
WD3 (auto, ~15 min)	Intake flow, validation and DQ refresh; critical errors -> rejection email	Power Automate, Dataflow	System
WD3-WD4	Correct data at source if rejected; re-extract as v2	SMART	OpCo Finance / SMART team
WD4 10:00	Missing-file check; alert if no file received (DQ01)	SCPL-03	System -> Admin
WD4-WD5	Admin reviews DQ page (checklist 9.3) and approves in Teams / Outlook	Approvals	Administrator
WD5 (auto)	Publish flag, archive, dataset refresh, personalised email	Power Automate,	System

When	Activity	Tool	Responsible
		Power BI	
WD5 onward	Users analyse in the app; questions to Admin via the Teams channel	Power BI App	Viewers
Daily	Refresh monitoring (and optional daily refresh of Actuals, Q08)	SCPL-03	System

9.2 RACI

Activity	OpCo Finance	SMART / IT	Administrator	BI CoE	Viewers
Close and post SC P&L in SMART	R/A	I	I	-	-
Schedule and deliver the monthly extract	C	R/A	I	C	-
Automated validation and refresh	-	I	A	R	-
Correct rejected data	R	R	A	I	-
Approve monthly publication	C	-	R/A	I	I
User and access management	-	-	R/A	C	I
Master-data mapping (OpCo, account, KPI)	C	C	R/A	C	-
Model / report changes (releases)	-	-	A	R	C
Use the app and give feedback	-	-	I	-	R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

9.3 Validation checklist before approval

Rule	Check	Error type	Severity	Where	If failed
DQ01	File present for reporting month in Inbox by WD+3 10:00	Missing file	Critical	Power Automate (scheduled check)	Stop; alert Admin + SMART team
DQ02	File type .xlsx/.csv and name SCPL_YYYYMM_vN	Wrong format	Critical	Power Automate + Power Query	Reject file to 99_Rejected
DQ03	All mandatory columns present with expected names	Missing columns	Critical	Power Query (fxCheckColumns)	Reject file
DQ04	Mandatory fields not blank (Year, Month, Country, OpCo, SC Function, Account, Scenario, Currency, Amount)	Missing OpCo/country/month	Critical	Power Query	Reject rows; block publish if > 0
DQ05	Year/Month valid and equal to the file period (all scenarios - one month per file)	Invalid period	Critical	Power Query	Reject file
DQ06	OpCo/Country exists in Dim_OpCo; Account exists in Dim_Account; Function in Dim_SC_Function	Unmapped master data	High	Power Query (anti-join)	Hold; Admin maps new item
DQ07	Scenario in {Actual, Budget, LY}	Invalid value	Critical	Power Query	Reject rows
DQ08	Currency = USD	Invalid currency	Critical	Power Query	Reject rows
DQ09	Amount is numeric; Pairs are non-negative integers	Invalid financial value	Critical	Power Query (try ... otherwise)	Reject rows
DQ10	No duplicate key (Period, OpCo, Function, Account, Scenario, Channel, AccDetail)	Duplicates	High	Power Query (Group By)	Block publish
DQ11	All expected OpCos loaded (vs Dim_OpCo active list)	Completeness	High	DAX / Power Query	Warn; publish with disclaimer if approved
DQ12	T accounts reconcile to sum of I accounts (tolerance USD 1)	Reconciliation	Medium	DAX (DQ page)	Warn
DQ13	Month-on-month movement > 50% on key lines	Outlier	Low	DAX (DQ page)	Info for review
DQ14	Dataset refresh succeeded	Refresh failure	Critical	Power Automate (refresh history API)	Alert Admin; retry once
DQ15	KPI measures return no error / divide-by-zero handled	KPI calculation error	Medium	DAX (DQ page, IFERROR/DIVIDE)	Warn

Administrator's additional sign-off checks (about 30 minutes):

- Group Net Turnover SC and Total Gross Profit agree to the SMART standard P&L report (tolerance USD 1K).

- Spot-check three OpCos (one per region) on Detailed P&L against SMART.
- Review month-on-month movements > 50% (DQ13) and new or unmapped master data (DQ06).
- Confirm that the email highlight figures match the Summary page, then approve.

10. Automation Plan

Component	Tool	Trigger	Automates
Extract	SMART scheduled report / job	Month-end close complete (WD3 08:00)	Export of the SC P&L to the SharePoint / OneDrive folder
Landing zone	SharePoint Online library SC_PL (01_Inbox, 02_Processed, 03_Archive, 99_Rejected)	-	Versioned, secured, auditable file store and history
SCPL-01 Intake & validation	Power Automate cloud flow	When a file is created in 01_Inbox	Name / format check, dataflow refresh, DQ query, reject or approval request, publish flag, archive, dataset refresh
Staging & validation	Power BI Dataflow (Power Query online)	Called by SCPL-01	Rules DQ02-DQ11, cleansing, mapping, unpivot
Semantic model refresh	Power BI Service (API refresh + scheduled fallback WD5 12:00)	After approval	KPIs, RLS, history
SCPL-02 Notification	Power Automate + Outlook (shared mailbox) + 'Run a query against a dataset'	Refresh succeeded and period Published	Personalised monthly email with highlights for the user's scope
SCPL-03 Monitor	Power Automate + Power BI REST API (refresh history)	Daily 07:00 and WD4 10:00	Refresh-failure alert, one retry, missing-file alert, refresh log
SCPL-04 Access sync	Power Automate + Entra ID connector	User_Access item created / modified / deleted	Group membership (app audience) and welcome email
Release management	Power BI deployment pipelines (Dev > Test > Prod)	Change request	Controlled changes to the model and report

10.1 Flow SCPL-01 - intake, validation and approval (steps)

- Trigger: SharePoint 'When a file is created in a folder' (SC_PL/01_Inbox).
- Condition: name matches SCPL_YYYYMM_vN and extension is .xlsx / .csv. If not, move the file to 99_Rejected and email the SMART team and Admin (DQ02).
- Refresh dataflow SC_PL_Staging, wait for completion, then refresh the model (DQ tables only for unpublished periods).
- 'Run a query against a dataset': count Critical and Warning exceptions, OpCos loaded and rows for the file period.
- Critical > 0: move the file to 99_Rejected, email the rejection with the exception list (Excel attachment) to OpCo Finance / SMART and Admin. Stop.
- Otherwise 'Start and wait for an approval' to the Administrator, with the DQ summary and a link to the DQ page. On approval, set Publish_Control status = Published, move the file to 03_Archive and refresh the model. On rejection, move the file to 99_Rejected with the Admin's comment.

10.2 Sample monthly email (SCPL-02)

Subject: Global SC P&L - August 2026 results published | Refreshed 07-Sep-2026 10:07 UTC

To: each active user in User_Access (personalised to their scope) **From:** sc-finance-reporting@<company> (shared mailbox)

Dear <First name>,

The Supply Chain P&L for **August 2026** is now available in the Global SC P&L app for your scope: <Region / OpCo / Function>.

Key highlights (USD, SAMPLE DATA):

- Net Turnover SC: USD 90.5 M (-3.9% vs Budget, +8.4% vs LY); YTD USD 624.6 M
- Gross Profit: USD 4.42 M, Gross Margin 4.9% (Budget 4.9%)
- EBITDA: USD 0.62 M (-6.3% vs Budget)

- Total Pairs: 7.56 M (-4.0% vs Budget); Freight cost per pair USD 0.121

- Alerts: 2 OpCo(s) with negative EBITDA, 7 with adverse PPV > 5%

Open the app: <<https://app.powerbi.com/groups/me/apps/...>> (detailed analysis stays in the app)

Data refreshed: 07-Sep-2026 10:07 UTC | Source: SMART ERP | Validation: PASSED (2 warnings reviewed by SC Finance)

Data-quality disclaimer: Figures are management information extracted from SMART after automated validation and Administrator approval. They may differ from statutory results and may be restated if late adjustments are posted. Data for ZM01 (Zambia) was not available this month. Please do not forward outside the authorised distribution.

Global Supply Chain Finance

Security note: highlights are computed per recipient. The flow reads the user's scope from User_Access and passes it as TREATAS filters in the DAX query, so no email shows data outside the user's RLS scope. Power BI subscriptions are the no-code alternative.

11. Error Handling and the Data Quality Page

Error	Detection	Where reported	Automatic action	Owner
Missing file	SCPL-03: no SCPL_<expected period> file by WD4 10:00 (DQ01)	Email / Teams; DQ page tile 'Reporting month'	Reminder to SMART team; escalation to Admin after 4 h	SMART / IT
Wrong format / name	SCPL-01 name / extension check; unreadable file (DQ02)	Rejection email; exceptions log	File -> 99_Rejected	SMART / IT
Missing columns	fxMissingColumns (DQ03)	Exceptions log; rejection email	File rejected, not loaded	SMART / IT
Duplicates	Group By on the business key (DQ10)	Exceptions log	Publication blocked until resolved	OpCo Finance
Missing OpCo / country / month	Mandatory-field rule (DQ04); active OpCo without rows (DQ11)	Exceptions log; OpCos loaded tile	Row rejected / warning for Admin approval	OpCo Finance
Invalid financial value	Non-numeric amount, negative or fractional pairs (DQ09)	Exceptions log	Row rejected (Critical)	OpCo Finance
Invalid currency	Currency <> USD (DQ08)	Exceptions log	Row rejected (Critical)	SMART / IT
Unmapped master data	OpCo / Account / Function not in dimensions (DQ06)	Exceptions log	Held; Admin maps the new item	Administrator
Refresh failure	Refresh history API (DQ14)	Teams / email; refresh history table	One automatic retry; last good data stays published	BI CoE / Admin
KPI calculation error	Blank ratio with non-blank base, reconciliation difference (DQ12, DQ15)	DQ page KPI checks	Warning; the KPI shows 'n.a', not an error	Administrator

Fail-safe principle: a failed file or refresh never overwrites the last published month. Users keep seeing validated data, and the DQ page and email show exactly what is missing.

11.1 Data Quality & Refresh Status page (Admin audience)

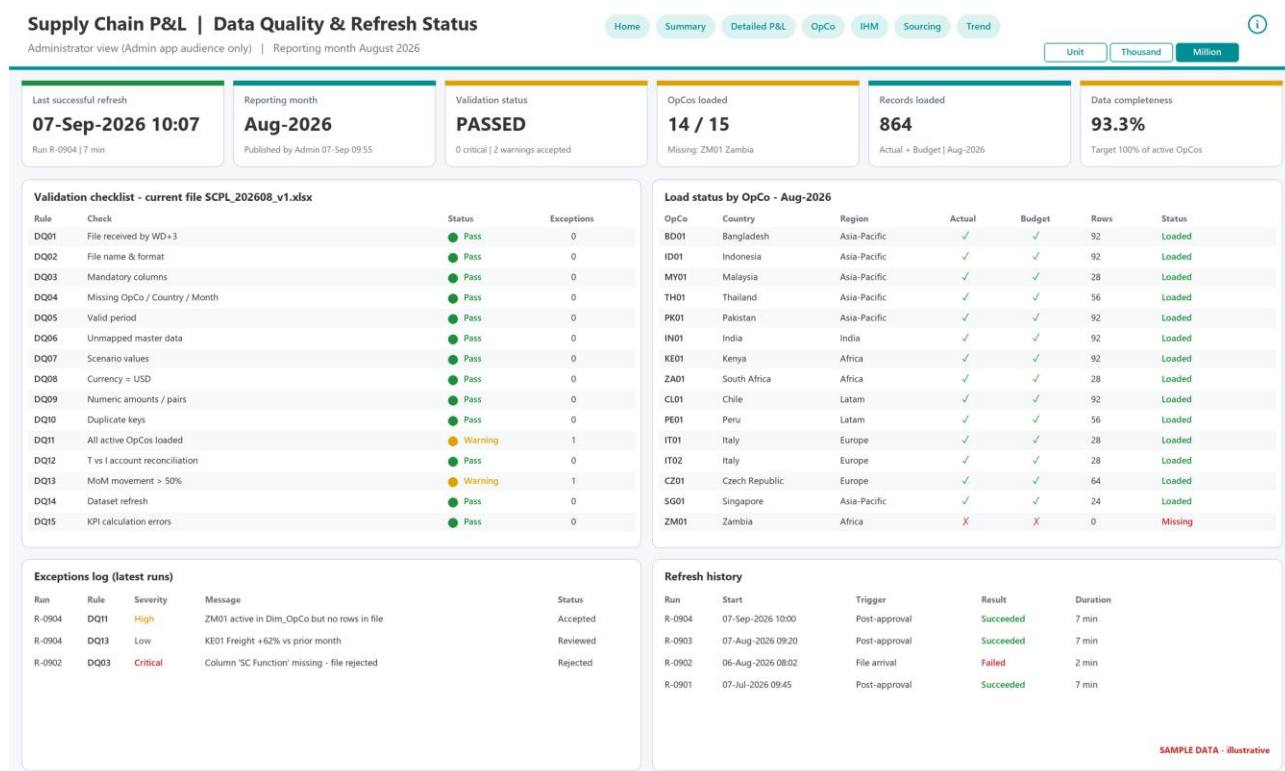


Figure 4 - Data Quality & Refresh Status page mock-up (SAMPLE DATA)

Zone	Content	Measures / source
Status tiles	Last successful refresh, Reporting month, Validation status, OpCos loaded (x / y), Records loaded, Data completeness %	[Last Refresh (UTC)], [Reporting Month], [Validation Status], [OpCos Loaded], [OpCos Expected], [Records Loaded (Latest)], [Data Completeness %]
Validation checklist	DQ01-DQ15, each with status light and exception count	DQ_Rules x DQ_Exceptions
Load status by OpCo	Actual / Budget received, rows, status (Loaded / Missing)	Dim_OpCo x Fact_SC_PL (+ unpublished staging)
Exceptions log	Run, rule, severity, message, file, row, status; export for OpCo Finance	DQ_Exceptions
Refresh history	Last runs, trigger, result, duration	Refresh_Log list (written by SCPL-03)
Reconciliation	T vs I account differences, SMART total checks	[Recon Diff Net Turnover], [Recon Diff Gross Profit]

12. Business Benefits

Time savings are estimates based on stated assumptions (A15, A16). They should be validated with the current actual effort during the pilot.

Activity	Today (hours / month)	With the app (hours / month)	Saving
Prepare country packs (22 countries x 1.5 h)	33	0 (automated)	33
Consolidation, checks and reconciliation	12	2 (DQ page review and approval)	10
Distribution emails and follow-up questions	8	2 (exceptions and user queries)	6
Access management	-	1	-1
Global SC Finance total	53	5	48 h / month (~576 h / year, ~0.3 FTE)
SPOC time spent requesting or looking for data (30 SPOCs x 1 h)	30	~5	~25 h / month

Benefit	Measure / indicator
Faster information	Publication moves from ~WD8 to WD5 (3 working days earlier); email on the same day
One version of the truth	One KPI catalogue and layout for 14+ OpCos; no reconciliation debates
Better decisions	Traffic-light exceptions (freight, PPV, recovery, margin) visible to the owner the same month
Control and audit	RLS, approval log, file archive and exception log replace emailed spreadsheets
Data quality	15 automated checks before every publication; visible completeness
Scalability	New OpCo = one master-data row; new KPI = one measure plus a Dim_KPI row; no redesign
Foundation for analytics	Clean monthly history in a governed model enables forecasting and AI (Deliverable 13)
Cost-saving opportunities	Benchmarking cost per pair across OpCos shows the gap to best quartile (see roadmap)

13. Future Roadmap

Phase	Capability	Approach / tools	Value
1 (0-3 months) MVP	Pilot with 3 OpCos, then roll out to all; mobile layout; subscriptions	This design	Quick win; adoption
2 (3-6 months)	Automated variance analysis	Key Influencers / decomposition tree; DAX price-volume-mix bridge (Turnover = pairs x price; cost per pair x pairs)	Explains 'why' automatically
2	AI-generated commentary	Copilot in Power BI (narrative visual) or Azure OpenAI in the SCPL-02 flow, grounded on the DAX query results	Monthly commentary draft in seconds
2	Anomaly detection	Built-in anomaly detection on trend lines; Fabric Data Activator alerts on thresholds	Early warning on freight / PPV spikes
2	Natural-language Q&A	Copilot / Q&A with a curated linguistic schema (synonyms: 'freight per pair', 'IHM')	Self-service for non-analysts
3 (6-12 months)	Forecasting	Fabric notebooks (Prophet / AutoML) on monthly history; new Scenario 'Forecast' in Dim_Scenario	Rolling forecast vs Budget
3	Savings identification	Gap to best-quartile cost per pair x pairs = savings opportunity (USD) per OpCo and cost category	Quantified cost-saving pipeline
3	Direct integration	SMART to Fabric Lakehouse (pipeline / API) instead of files; daily actuals	Near real-time, less file handling
3	Extended scope	Local currency and FX analysis, additional KPIs (inventory, service level), write-back commentary (Power Apps)	Richer management reporting

14. Assumptions & Open Questions Log

14.1 Assumptions

ID	Area	Assumption	Impact if wrong
A01	Extract	SMART can deliver a scheduled monthly long-format file SCPL_YYYYMM_vN (.xlsx / .csv) with the columns in 4.4; one reporting month per file (Actual, Budget and LY rows for that month).	Power Query mapping changes
A02	Extract	SC Function (Imports, Local Sourcing, Manufacturing, Global Sourcing) and BU are available on every row.	Manufacturing / Procurement KPIs cannot be split
A03	LY	LY uses the SMART 'LY' scenario when supplied, otherwise prior-year Actual from history.	Different LY at FX / restatement level
A04	Signs / currency	Amounts are in USD as delivered by SMART; revenue and credits positive, costs negative. Currency column = USD.	Sign flip needed via Dim_Account
A05	KPI	Channel Cost % = Channel Expenses / Net Turnover SC (template 'Channel Expenses %').	Denominator change
A06	KPI	'Turnover' in Cost of Sales % = Net Turnover SC.	Ratio level changes
A07	KPI	EBITDA = Channel Operating Profit (SMART PL_CP0000T) + Enabling Functions Expenses; the template sample equals GP + Channel Expenses because the other lines are blank.	EBITDA level
A08	KPI	EBIT = EBIT BNRI + Non Recurring Items (signed as in SMART); EBIT BNRI = EBITDA as	EBIT level / sign

ID	Area	Assumption	Impact if wrong
		in the template.	
A09	Layout	Summary card allocation per section as in 7.2 (the template shows only 'KPI 1').	Card content swap (no redesign)
A10	KPI	Average Cost per Pair - LS denominator = LS Purchase Pairs.	Ratio level
A11	Alerts	GP margin: Actual vs Budget or LY < -1 pp; Freight: cost per pair vs LY > +10%; PPV: adverse vs Budget > 5%; Recovery: vs Budget > 10%; EBITDA < 0.	Different flags
A12	Calendar	Financial year = calendar year (Jan-Dec), as in the template month lists.	YTD logic (fiscal offset)
A13	Master data	BU is an attribute of OpCo (AW Lab is a separate OpCo).	BU would move to the fact
A14	Tooling	Microsoft 365 with SharePoint, Power Automate (standard + premium HTTP), Power BI Pro; Entra ID groups.	Licensing / design of flows
A15	Benefits	Current effort about 53 h per month, publication about WD8.	Benefit figures
A16	Benefits	30 SPOCs each spend about 1 h per month requesting or looking for data.	Benefit figures
A17	Accounts	TBC account codes (Q07) are loaded but not used in KPIs until confirmed.	Some P&L rows blank
A18	Data	All values in the mock-ups, sample workbook and email are SAMPLE / dummy data.	-

14.2 Open questions

ID	Question	Owner	Needed by
Q01	Can SMART schedule the extract to SharePoint automatically? Exact column names and format?	SMART / IT	Build start
Q02	Is SC Function and BU on each row? Is 'Supply Chain' the total of the four functions?	SMART / Admin	Build start
Q03	LY: SMART scenario (restated) or prior-year Actual?	Admin	Build start
Q04	USD conversion rate basis (actual / budget / constant rate)?	Group Finance	Build start
Q05	Confirm the sign convention in the SMART extract.	SMART / IT	Build start
Q06	EBITDA / EBIT build-up: include Marketing, 4-Walls, Enabling Functions? Template EBIT = EBIT BNRI - NRI?	Admin / Group Controlling	KPI sign-off
Q07	Meaning of PL_210000T, PL_220000T, OVERHEADS, PL_370000I, 970000, PL_940000T, PL_950000T.	SMART / Controlling	KPI sign-off
Q08	Monthly publication only, or daily refresh of Actuals as the template note says?	Admin	Design sign-off
Q09	Licensing: Pro for all viewers or Premium / Fabric capacity? Number of viewers?	IT / BI CoE	Go-live
Q10	Confirm the 12 Summary cards and the traffic-light comparison bases.	Admin / management	Design sign-off
Q11	Denominators for Channel Cost % and Cost of Sales %.	Admin	KPI sign-off
Q12	Should the Quarterly period be quarter-to-date or full quarter?	Admin	Design sign-off
Q13	Are KPIs needed for Global Sourcing (none in the KPI file)?	Sourcing Head	Phase 2
Q14	Email: personalised highlights per user (proposed) or group level for management only?	Admin	Go-live
Q15	Is the 'Industrial' region an OpCo grouping or a function?	Admin	Build start

Appendix - companion files

File	Content
SC_PL_Prototype_Sample_Data.xlsx	SAMPLE data set (19,008 fact rows, Jan-2025 to Aug-2026) plus all dimension, security, DQ and control tables. Load into Power BI Desktop to build the prototype.
SCPL_202608_v1_SAMPLE.csv	One monthly extract in the proposed SMART format (drop into the Inbox to test the flow)
SC_PL_DAX_Measures.dax	Complete DAX library: measures, calculation groups, RLS, DQ measures, Tabular Editor script
SC_PL_PowerQuery_M.pq	Complete M code: parameters, functions, staging, validation, fact, dimensions
SC_PL_PowerBI_Theme.json	Report theme matching the template colours
images/	Architecture, star schema, Summary page and DQ page mock-ups